

Circular Motion

For motion in circles, we require angular displacement instead of normal displacement

$$\text{Angular displacement} = \Delta \theta \quad \text{rad}$$

$$\text{Angular velocity} = \overset{\text{omega}}{\omega} = \frac{\Delta \theta}{\Delta t} \quad \text{rad}\cdot\text{s}^{-1}$$

$$\text{Angular acceleration} = \alpha = \frac{\Delta \omega}{\Delta t} \quad \text{rad}\cdot\text{s}^{-2}$$

$\uparrow$ alpha

Two main motions

1. Constant angular velocity

Similar to kinematics : $v = \frac{\Delta x}{\Delta t}$

in circular motion we use : $\omega = \frac{\Delta \theta}{\Delta t}$

2. Constant angular acceleration

Circular Motion

$$\theta_f = \theta_o + \omega t$$

$$\omega_f = \omega_i + \alpha t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha \Delta \theta$$

$$\Delta \theta = \frac{1}{2}(\omega_i + \omega_f)t$$

$$\Delta \theta = \omega_i t + \frac{1}{2}\alpha t^2$$

Kinematics

$$x_f = x_o + vt$$

$$v = v + at$$

$$v^2 = u^2 + 2as$$

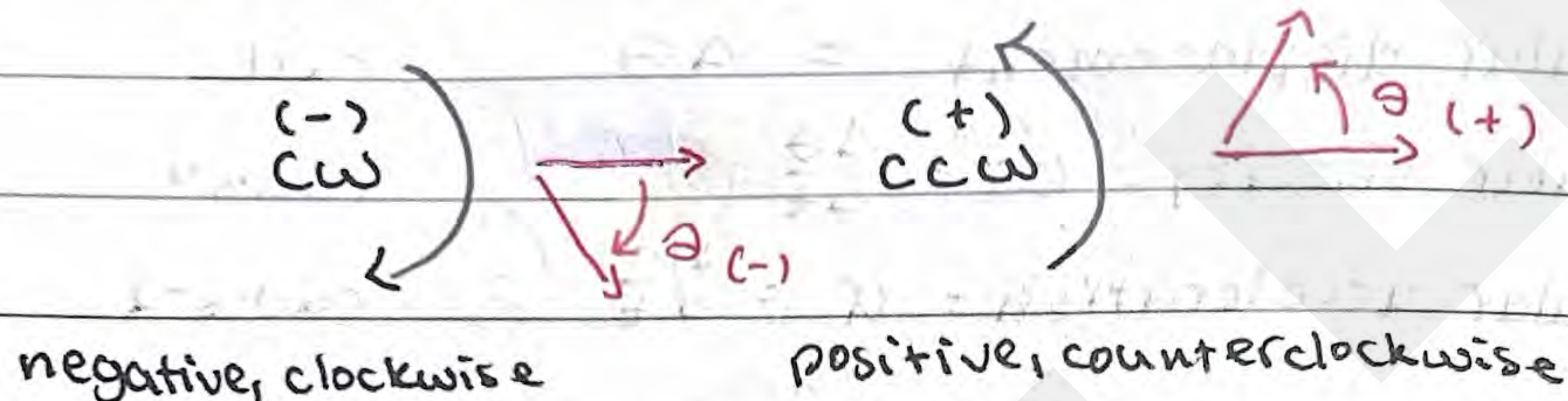
$$s = \frac{1}{2}(u+v)t$$

$$s = ut + \frac{1}{2}at^2$$

Circular motion

↙ in the 3rd dimension

Angles are vectors, the generally agreed upon norm for signs are...



Similar to kinematics, we can also describe circular motion with graphs

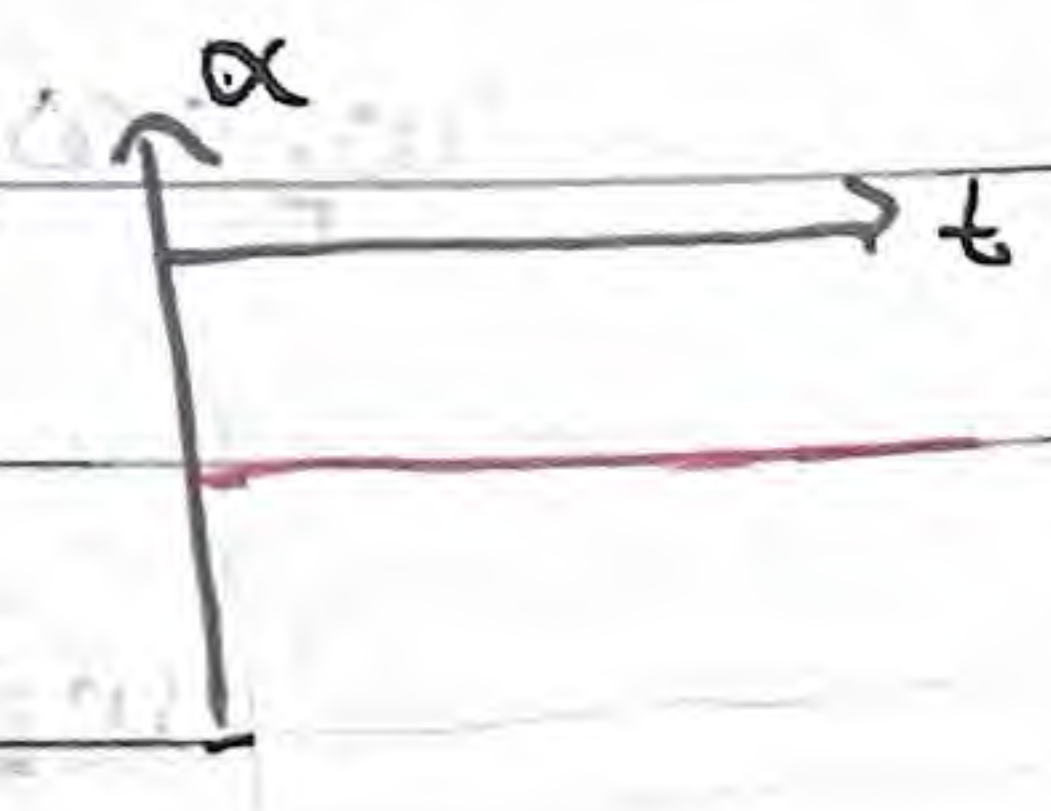
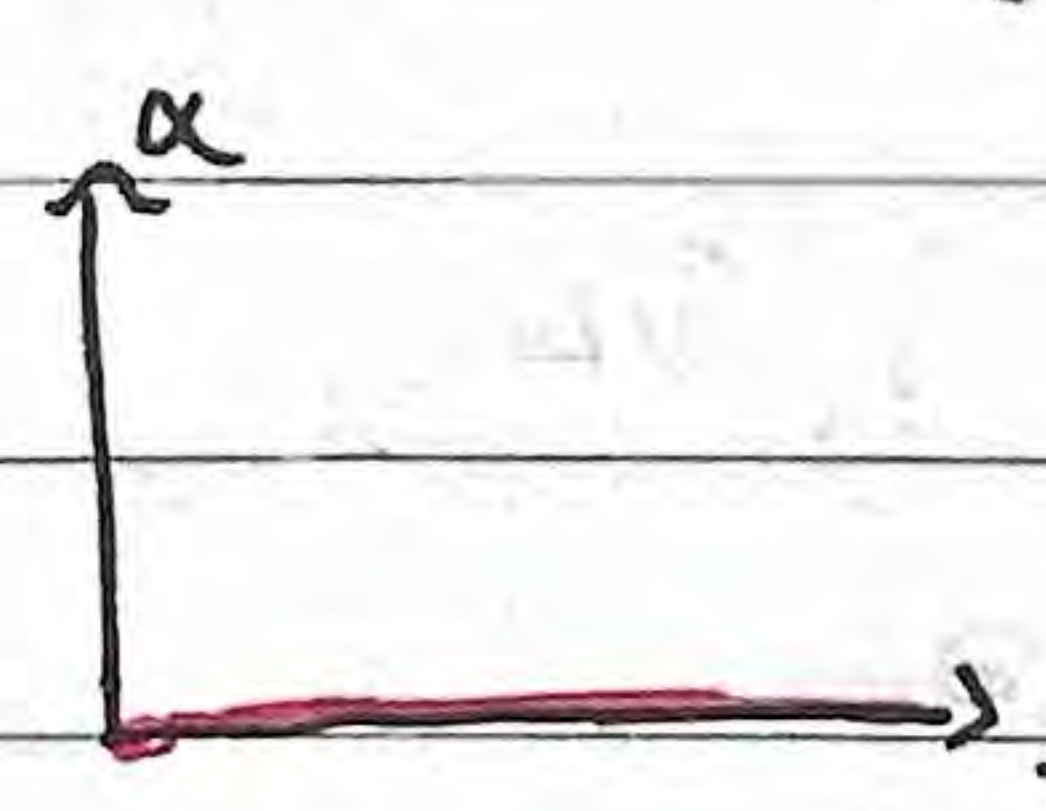
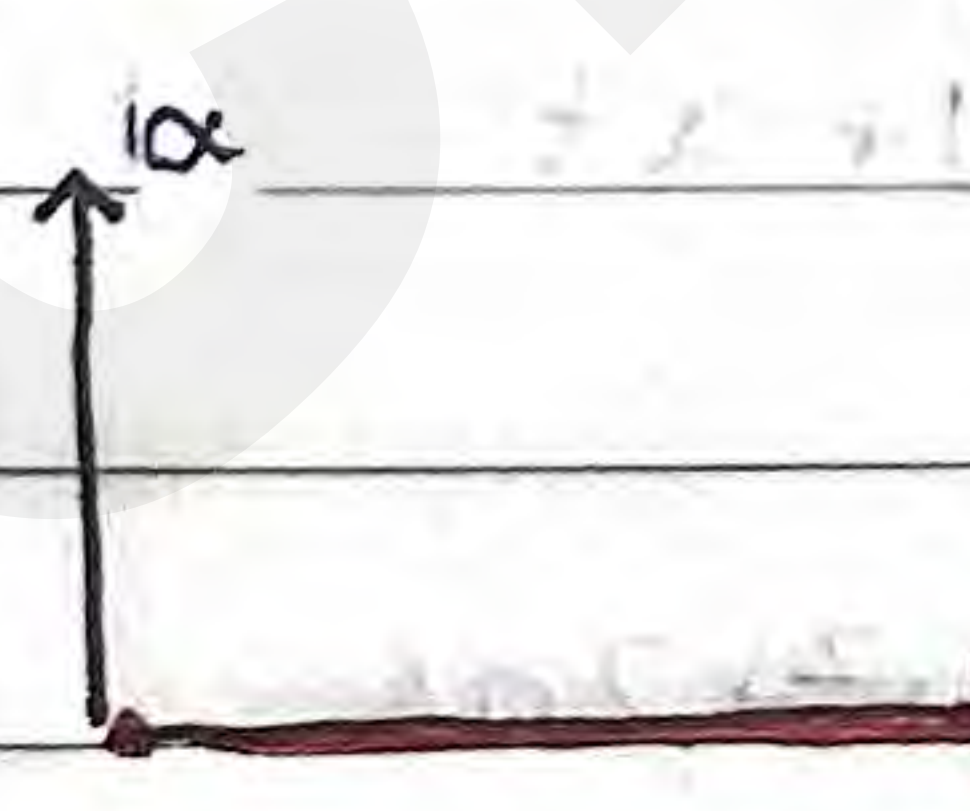
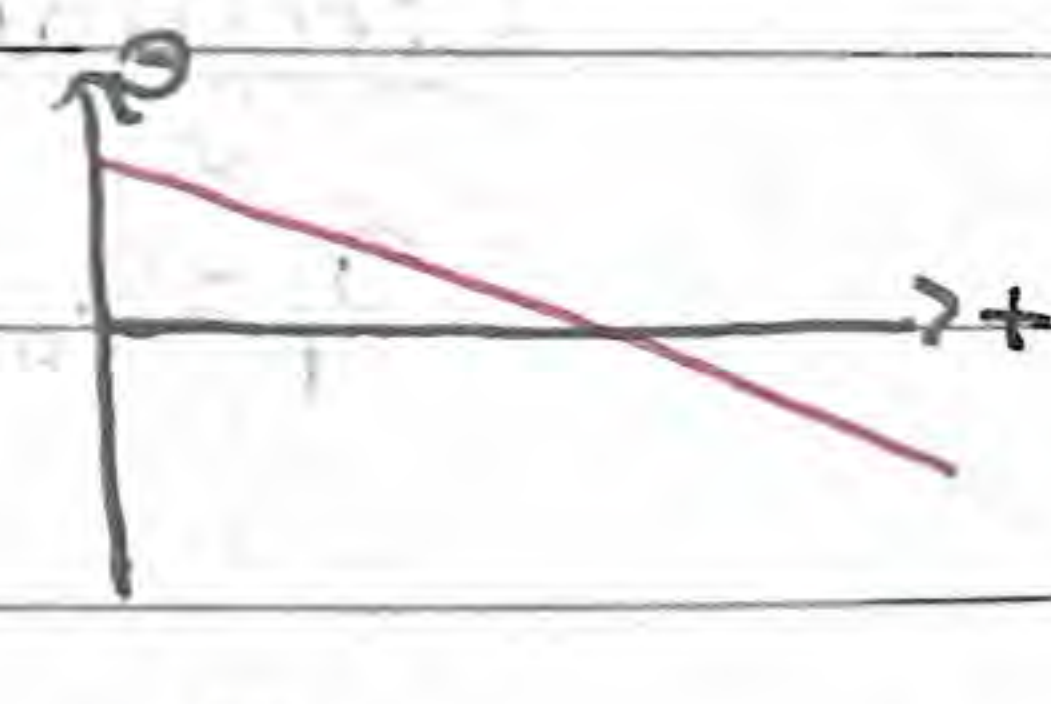
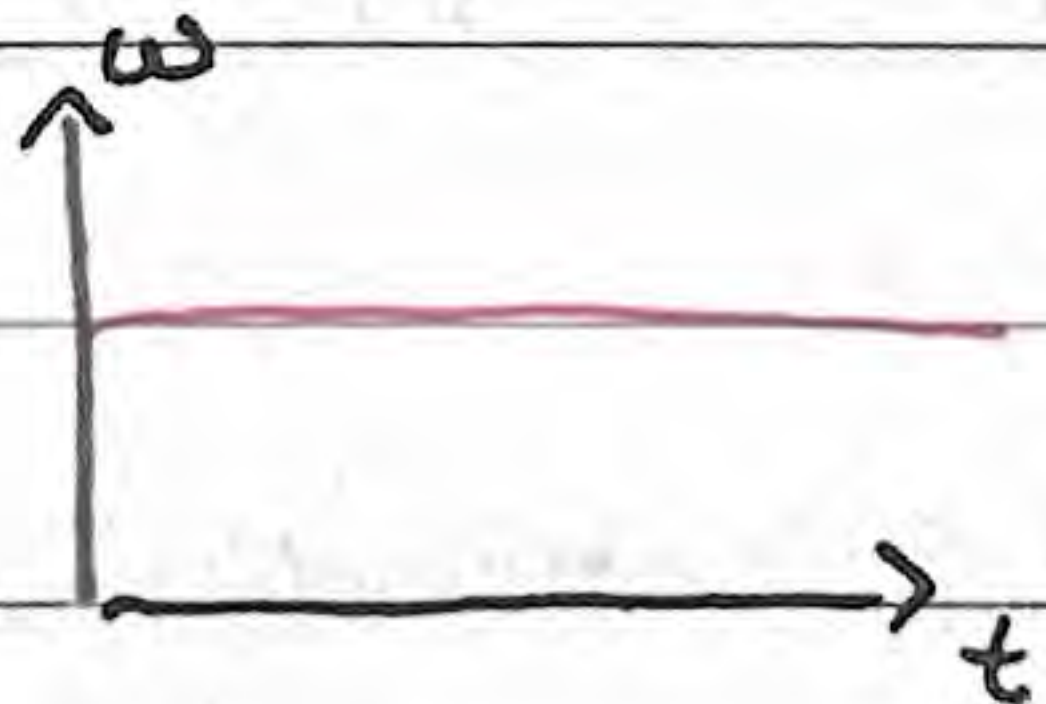
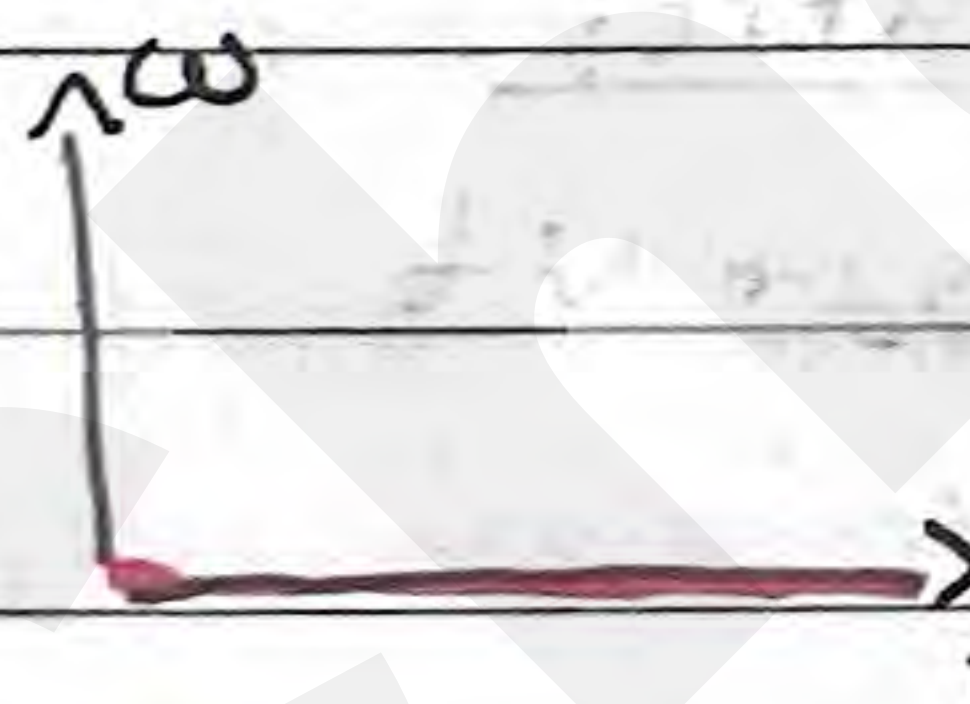
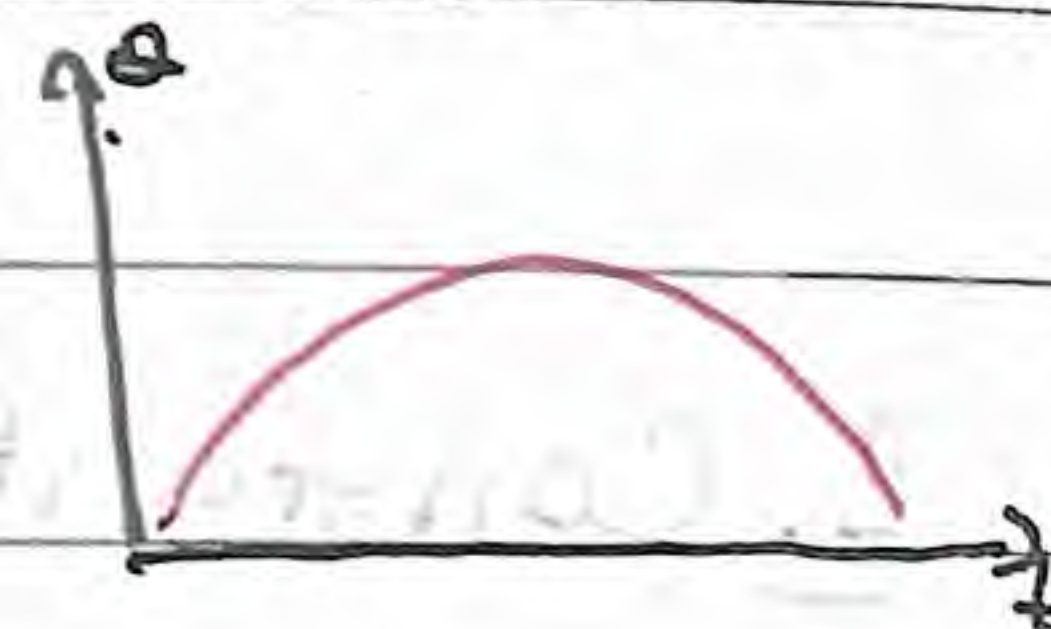
Constant angle



Constant velocity



Constant acceleration



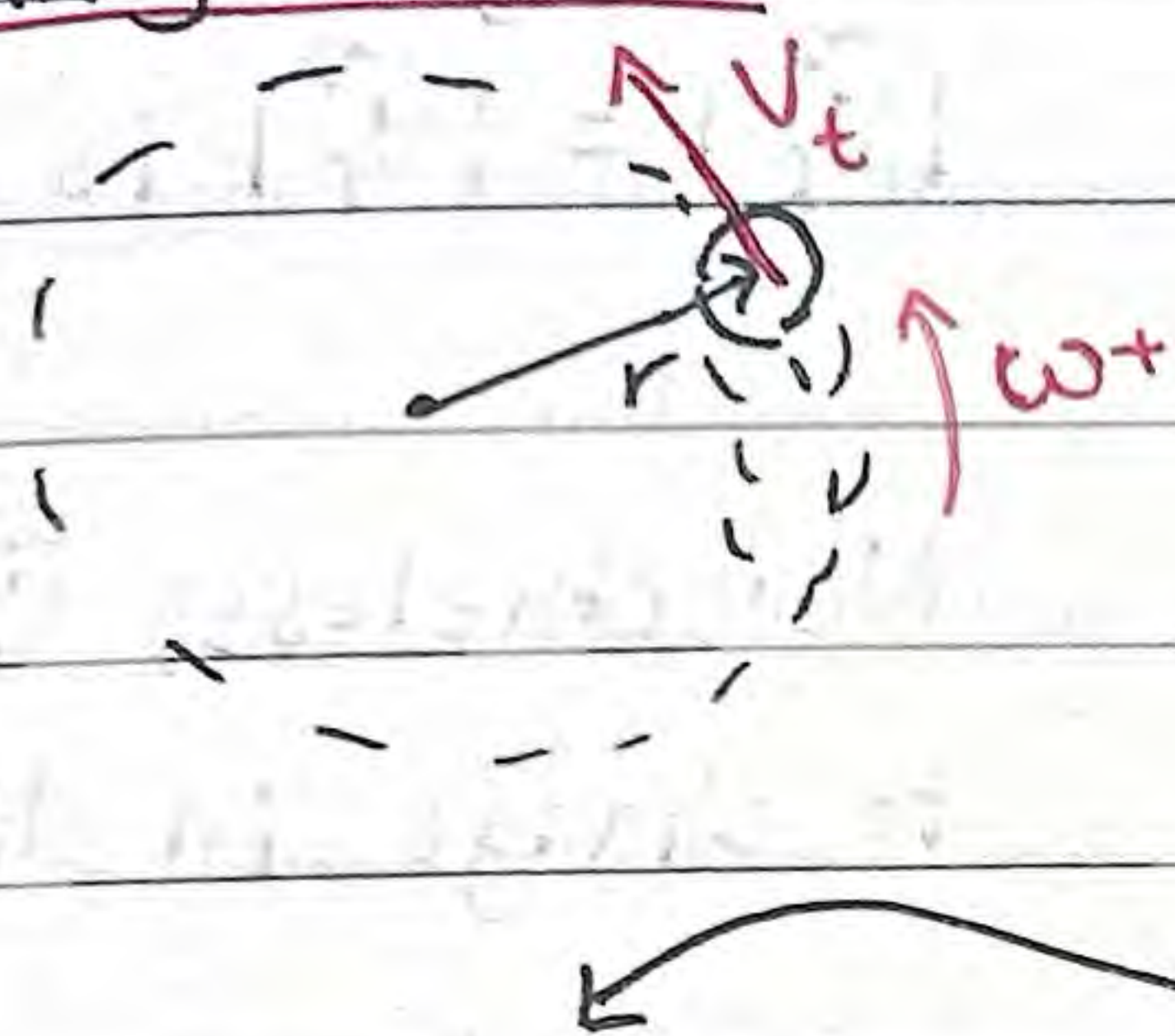
The area beneath $\omega(t)$ is θ displacement.

The slope of $\omega(t)$ is α at that instant.

Circular motion

Aside from angles, we also need to deal with two other types of quantities:

1. Tangential



Tangential Velocity

- Always perpendicular to radius
- Comes from definition of radian

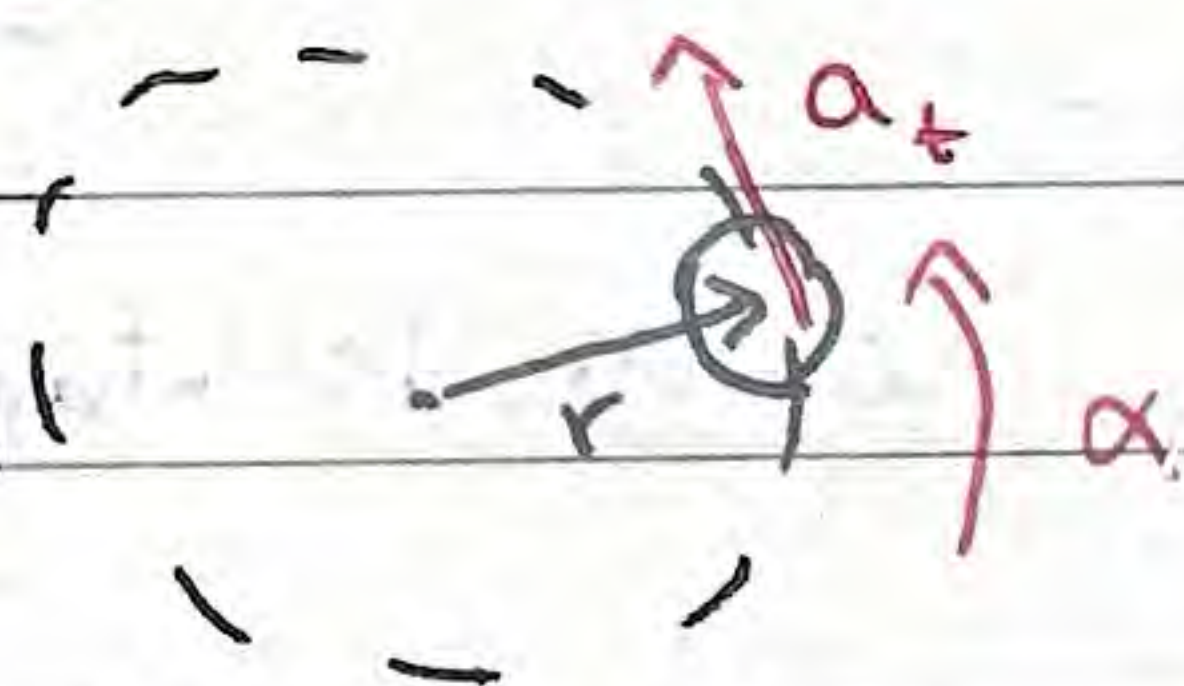
$$s = r \cdot \theta$$

$$\frac{d}{dt}[s] = \frac{d}{dt}[r \cdot \theta] \Rightarrow v_t = r \cdot \omega$$

Tangential Acceleration

When $v_t = r \cdot \omega$ is differentiated

$$a_t = r \cdot \alpha$$

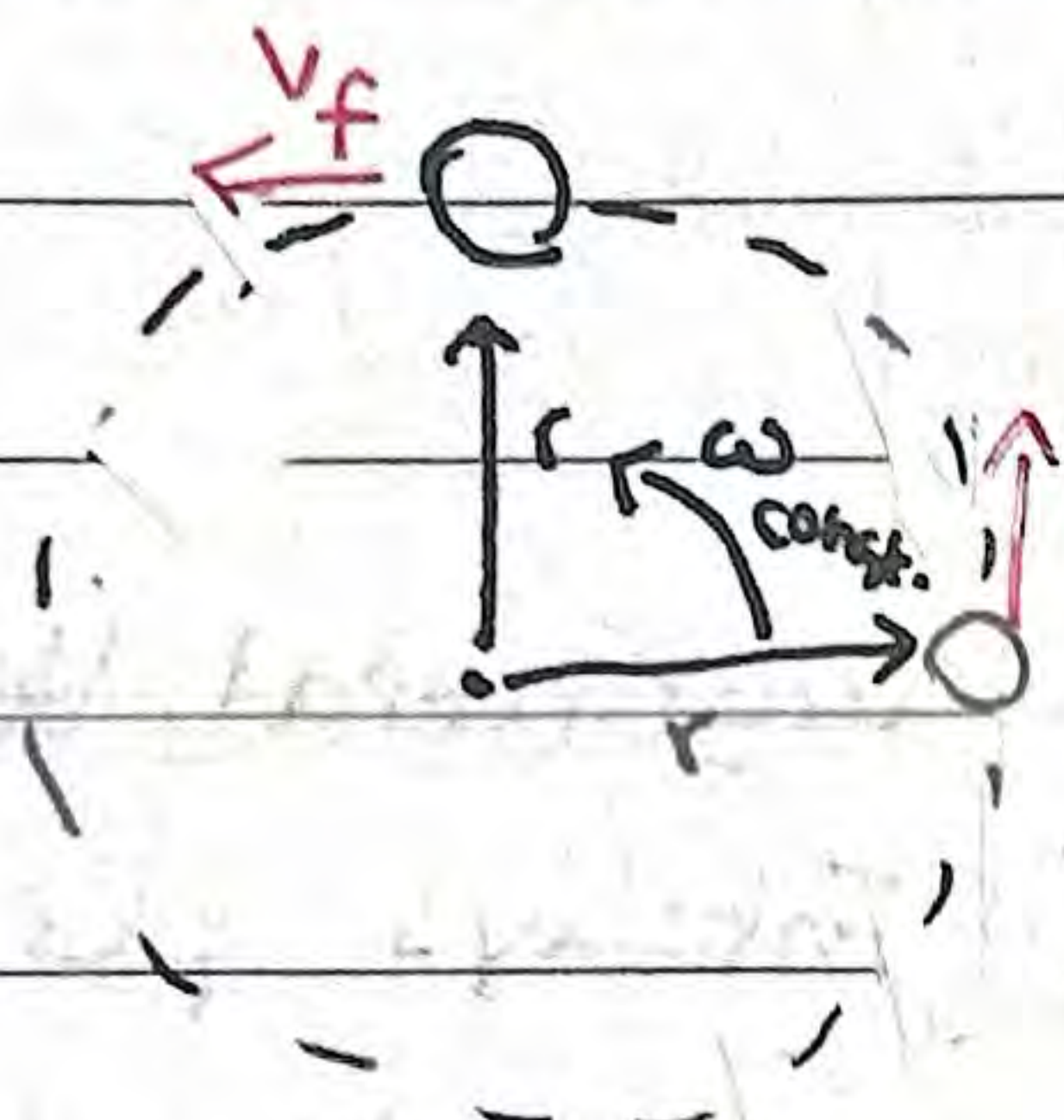


This is the tangential acceleration, always perpendicular to radius

Circular motion

2. Centripetal

consider the following scenario

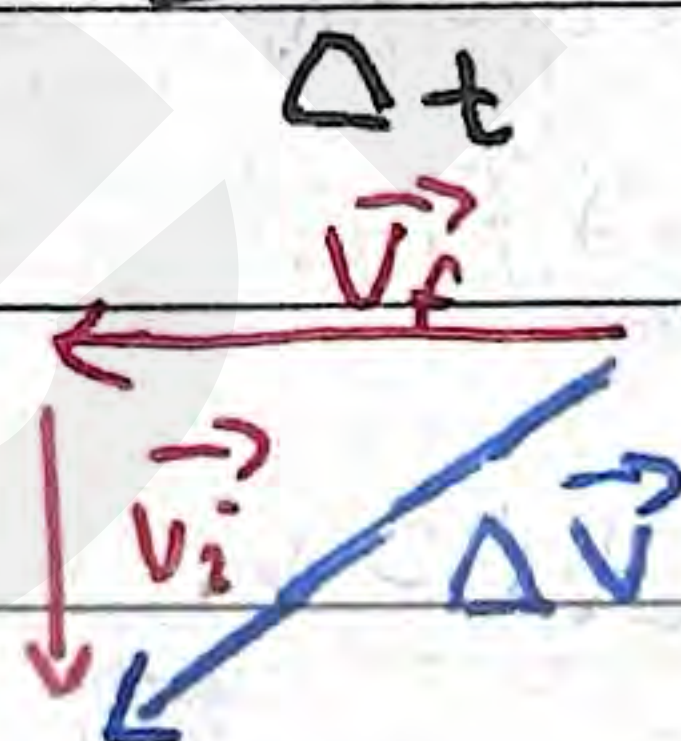


Assuming ω is constant,
 $|\vec{v}_f| = |\vec{v}_i|$ is also constant

Nevertheless, $\vec{v}_i \neq \vec{v}_f$ due
 to change in direction

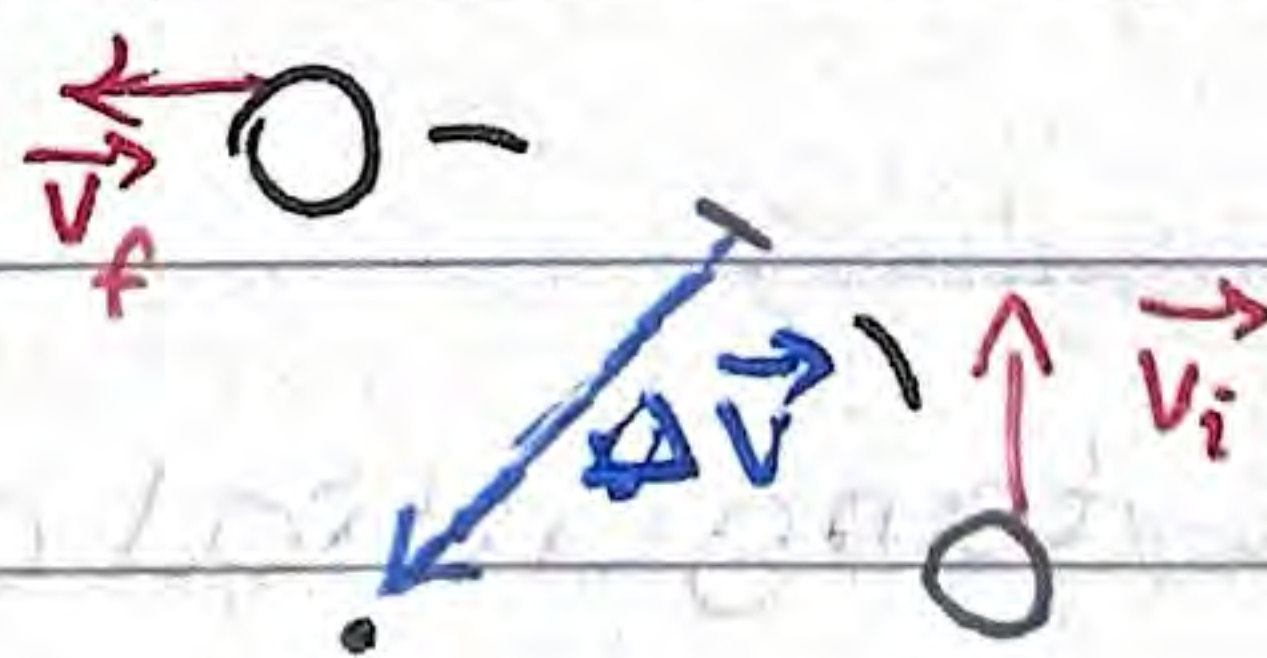
Thus there must be acceleration

$$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$$



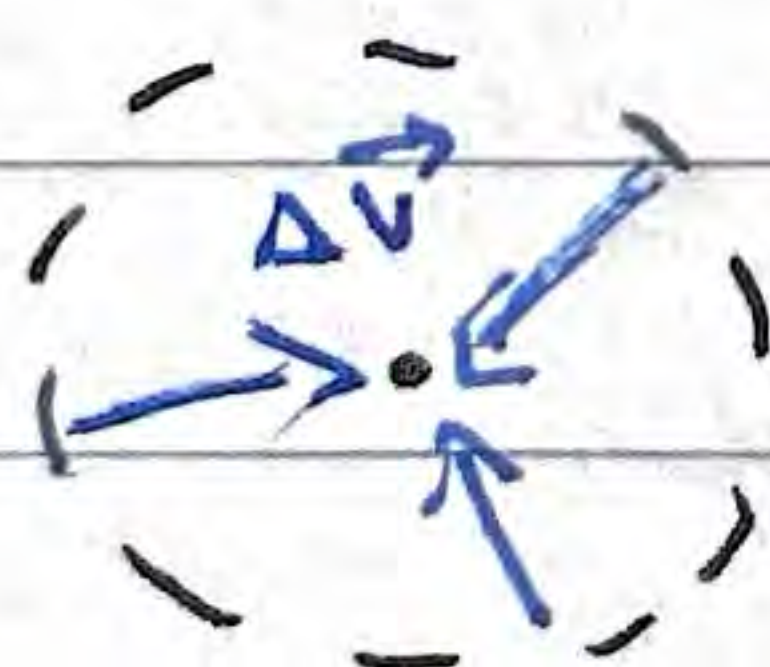
Let's dissect the
 vectors...

when looking at the diagram...



notice that
 $\Delta \vec{v}$ points
 inwards to
 the radius

Same will occur for other points



Circular motion

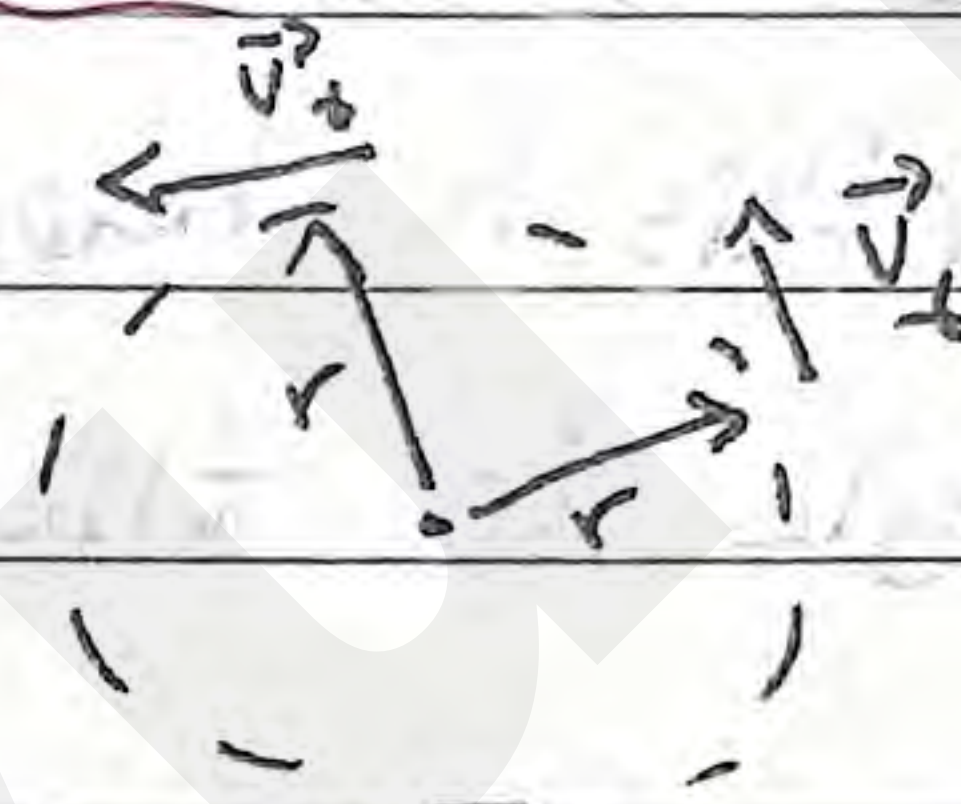
2. Centripetal

This change in velocity indicates an acceleration inwards.

Centripetal Acceleration

"center seeking"

where $a_c = \frac{|\vec{v}|^2}{r}$



for $|\vec{v}|$ is constant, r is radius

As such...

For any constant angular/tangential velocity of circular motion, there must still exist a centripetal acceleration

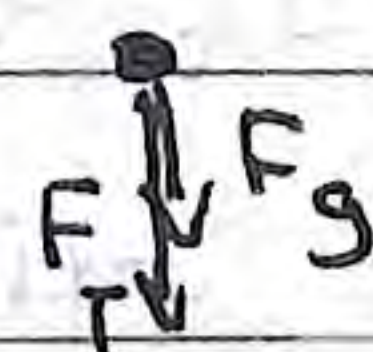
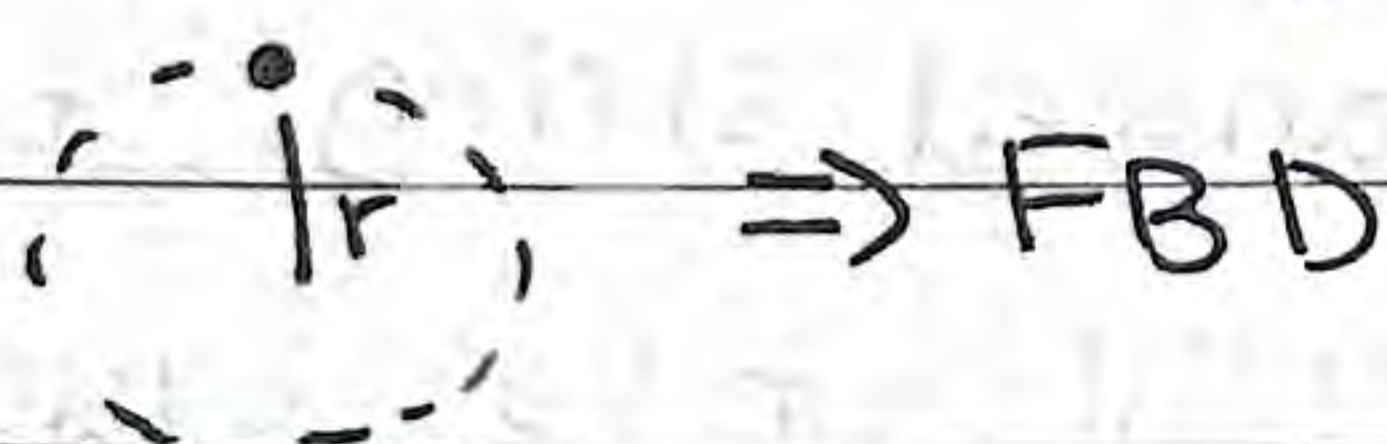
* Centripetal force does not exist! (the centripetal acceleration only occurs from OTHER forces)

Free body diagram of circular motion

The F_{net} will always be mass $\times$ centripetal acceleration

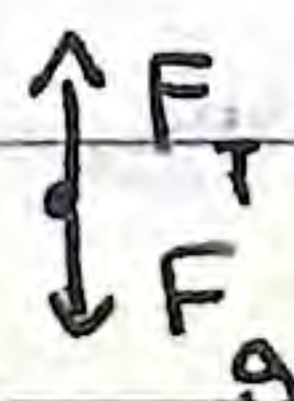
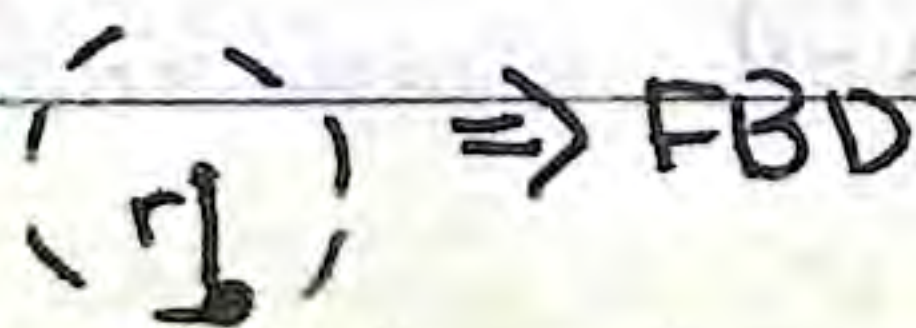
Diagram 1

vertically rotated



Thus $-m \cdot a_c = -F_T - F_g$

Diagram 2



Thus $m \cdot a_c = F_T - F_g$

tension greater when object at the bottom

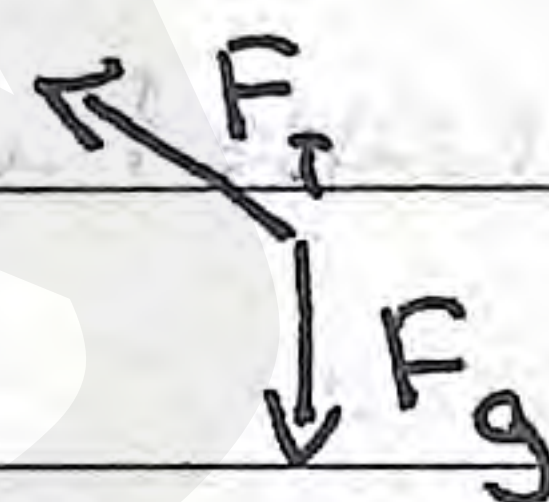
Circular Motion

Horizontal circular motion

← → An object can also be rotated in a horizontal circular motion

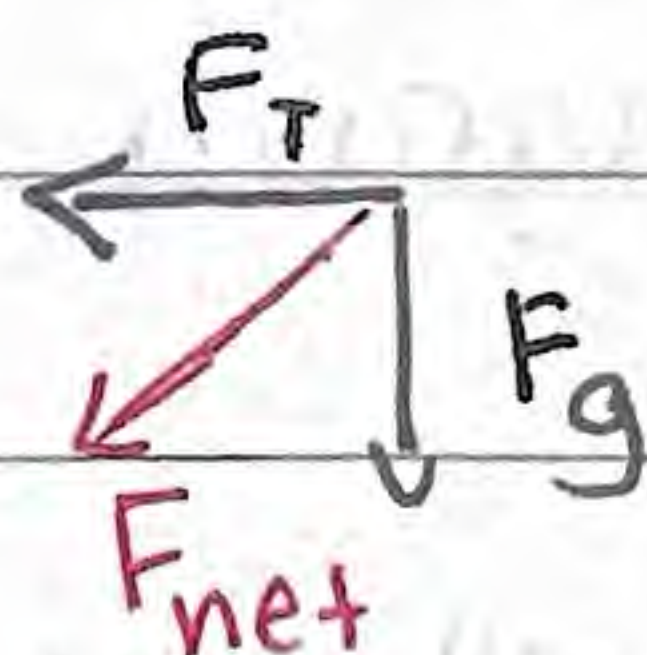
For this to happen, the string must be at an angle to the path...

This is because...



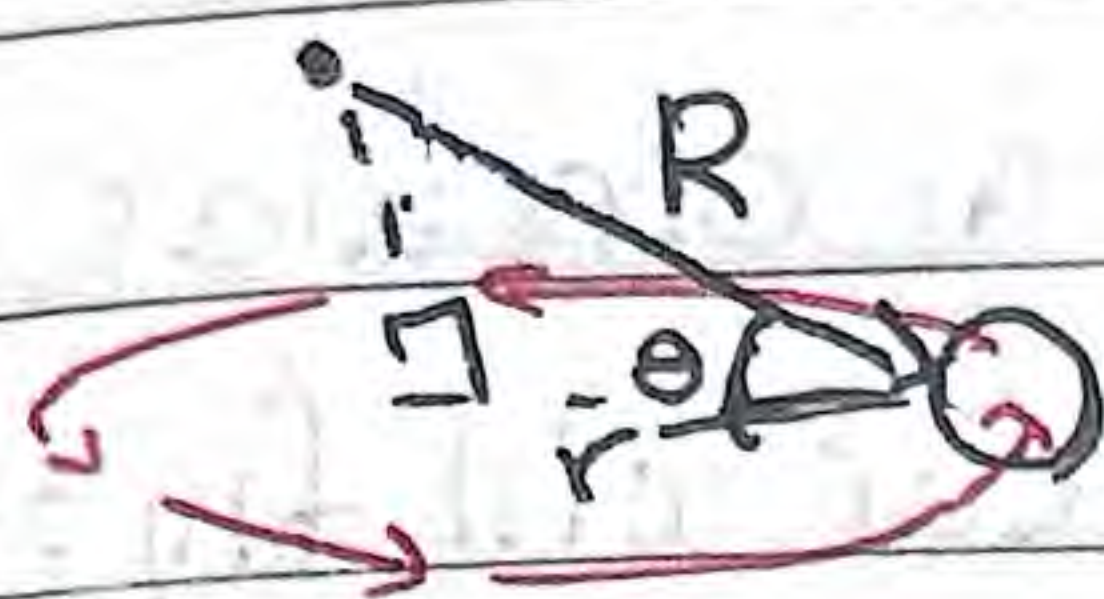
For there to be $F_{\text{net } y} = 0$, $F_{T_y} = F_g$ indicating string must be tilted at an angle

A scenario where the string is horizontal cannot occur because a downward acceleration will occur



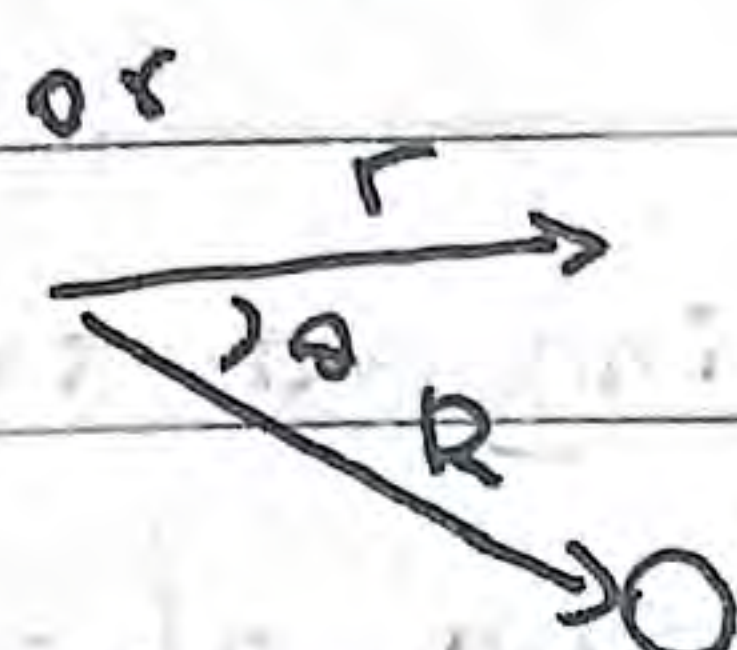
As a result of the angled string, the length of the string will not be equal to the radius of the motion

Circular Motion



Where...

$$r = R \cdot \cos \theta$$



But for this to work, a min. velocity is required.

$$\underline{x} : -F_{Tx} = -m \frac{v^2}{r}$$

* v is tangential velocity (horizontal)

$$F_T \cdot \cos \theta = m \frac{v^2}{r}$$

$$\underline{y} : F_{Ty} = F_g$$

$$F_T \sin \theta = mg$$

$$\therefore \tan \theta = \frac{gr}{v^2}$$

* Dividing F_{net_x} by F_{net_y} is common in circular motion questions (to cancel m, F_T etc.)

$$v = \sqrt{\frac{g \cdot r}{\tan \theta}}$$

Thus $v = \sqrt{g \cdot r}$

For vertical circular motion, a min. velocity is also expected

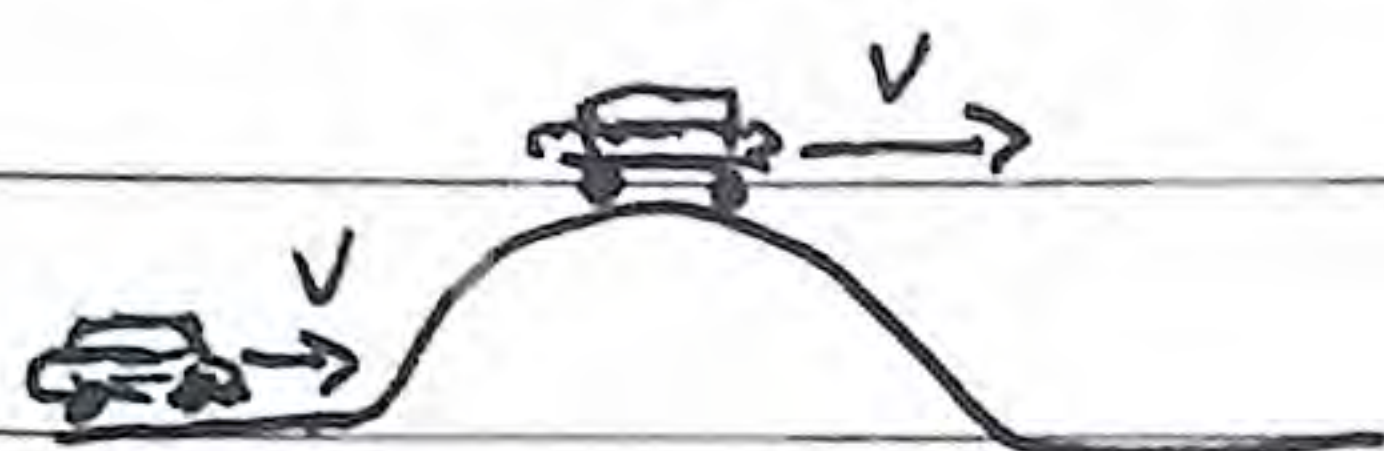
- Tension must always ≥ 0 ; $F_{Tmin} = 0$ and occurs at very top of the trajectory $\Rightarrow$ Same with other forces acting centripetally ($F_N = 0$ when no longer contact)

Circular motion

E.g. Consider a car climbing a circular hill

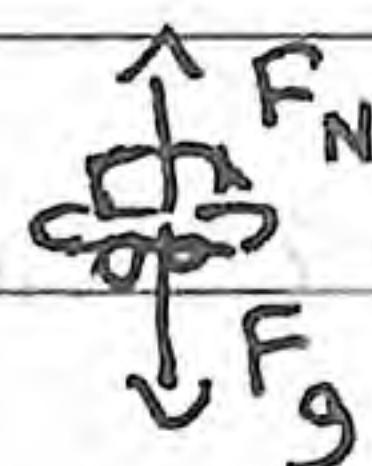
- At a certain speed v , the car will fly off.

So find v .



Looking at the free body diagram of the car

$$F_{\text{net } y} = m \cdot a_y$$



$$F_{\text{net } y} = F_N - F_g = -m \frac{v^2}{r}$$

The key to realize here is normal force only

exists when in contact.

Thus the max. velocity should be less than when

$$F_{\text{net } y} = -F_g = -m \frac{v^2}{r}$$

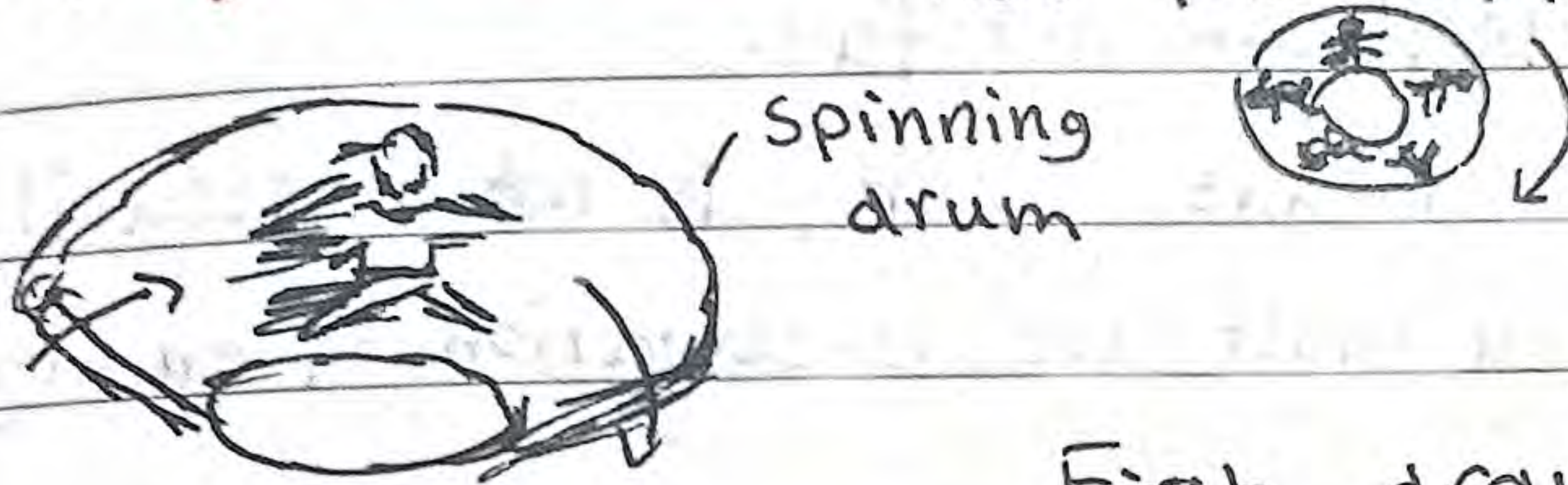
$$\Rightarrow mg = m \frac{v^2}{r}$$

$$\text{Thus } v < \sqrt{r \cdot g}$$

$$F_N = 0$$

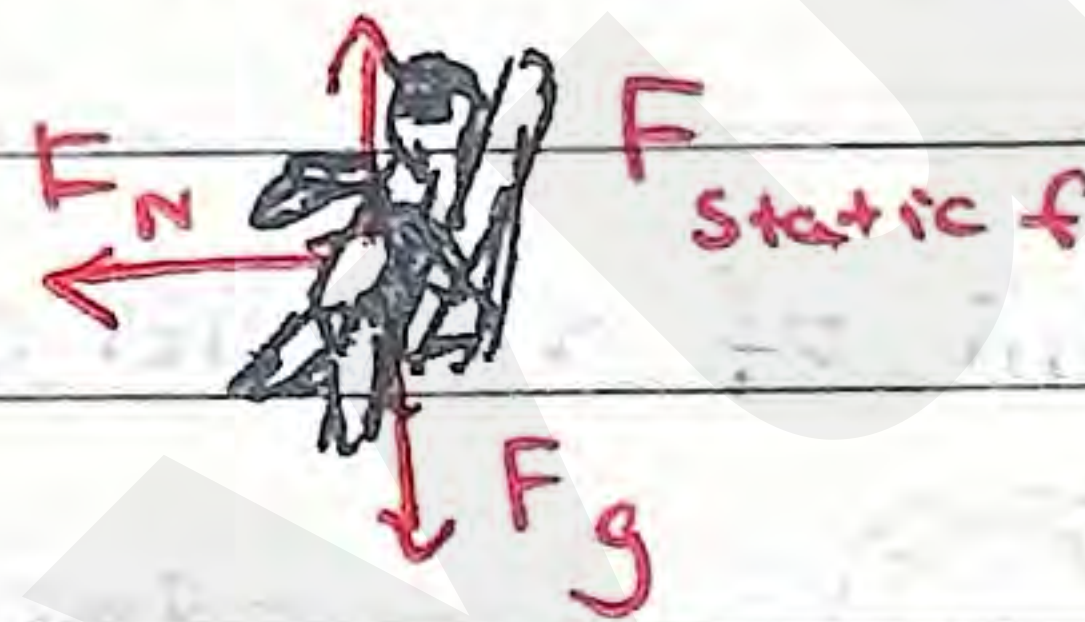
Circular motion

E.g. Drum ride (minimum speed v so the children do not fall off)



First, draw freebody diagram

* F_N acts as centripetal force here



$$F_{\text{net},x} = -F_N = -m \frac{v^2}{r}$$

$$F_{\text{net},y} = 0 = F_s - F_g$$

$$\frac{F_{\text{net},x}}{F_s} = \frac{F_N}{F_s} = \frac{v^2}{r \cdot g}$$

$$F_s = F_g = m \cdot g$$

$$v^2 = r \cdot g \cdot \frac{F_N}{F_s}$$

$$\text{since } F_s \leq \mu_s \cdot F_N$$

$$v^2 \geq r \cdot \frac{g}{\mu_s}$$

$$\therefore v \geq \sqrt{\frac{r \cdot g}{\mu_s}}$$

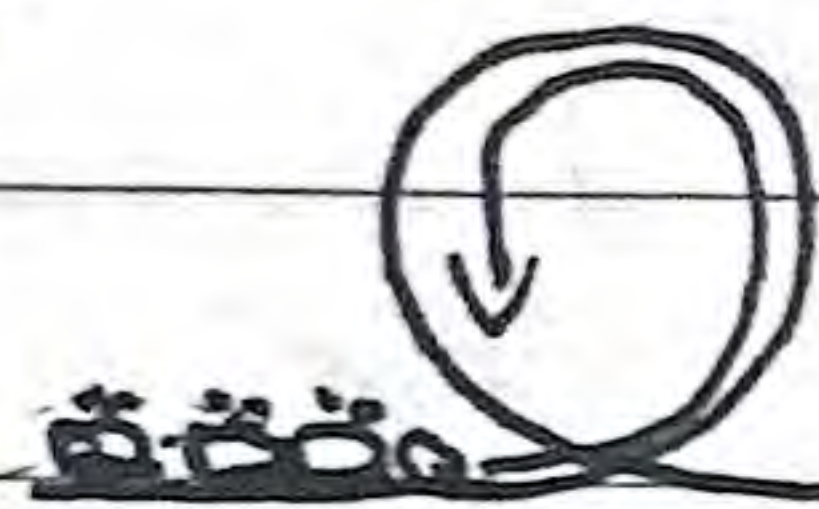
Circular motion and conservation of energy

Like done previously, circular motion requires a minimum velocity to not fall.

• But, when constant velocity is not necessarily guaranteed, you must use conservation of energy

Derived earlier, $v_{\min} = \sqrt{r \cdot g}$

E.g. Think of a roller coaster (nonconst. velocity)



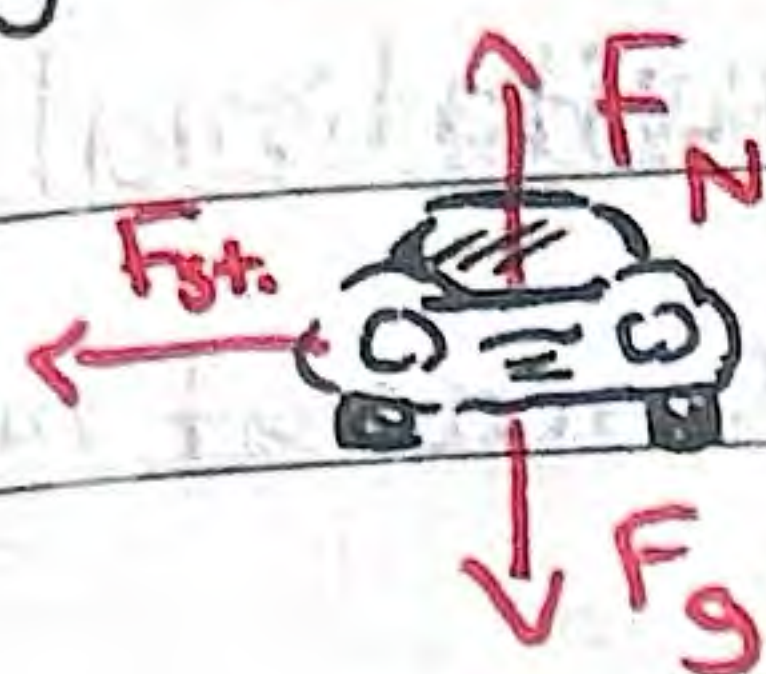
$v_{\min}$ occurs at the very top of the track

• Energy is no longer purely kinetic, some are converted into gravitational

Thus, let v_0 = initial speed potential energy

$$\frac{1}{2} m v_0^2 = m \cdot g \cdot h + \frac{1}{2} m v_{\min}^2$$

$$v_0 = \sqrt{5 \cdot r \cdot g}$$

Circular motionE.g. Car on a curve

y: $F_N = m \cdot g$

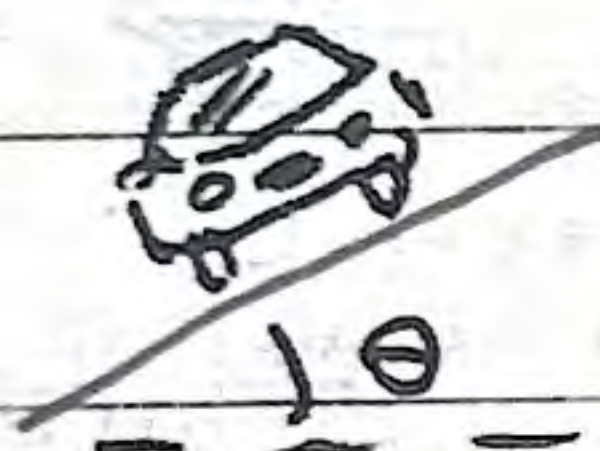
x: $m \frac{v^2}{r} = \mu_s \cdot F_N$

$$\frac{v^2}{r \cdot g} = \mu_s$$

$$\therefore v = \sqrt{r \cdot g \cdot \mu_s}$$

Banked curves

(planes, highways)



$$-F_N \sin \theta = -m \frac{v^2}{r}$$

$$F_N \cos \theta = m g$$

$$\therefore v = \sqrt{g \cdot r \cdot \tan \theta}$$

Gravitational Force

Force resulting from the interaction between two masses

- Similar to electric force, the gravitational force's direction will depend on the masses' center of mass



* Gravitational forces will always be attractive

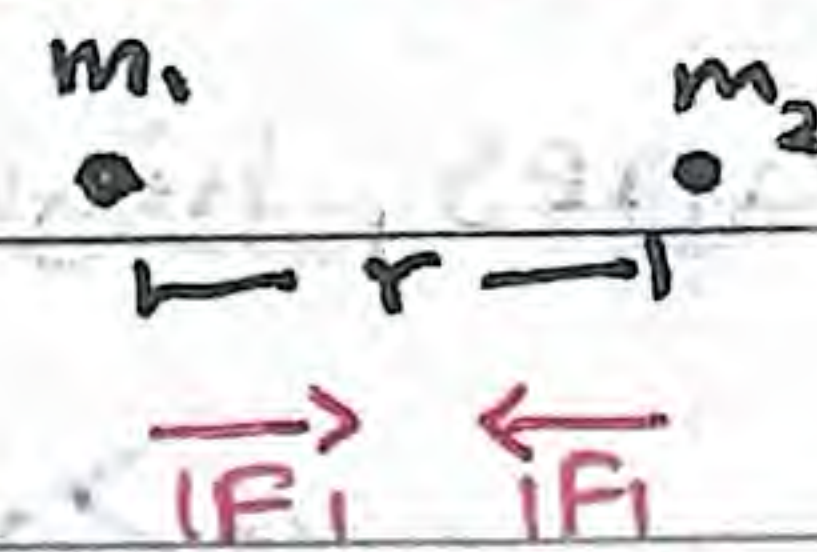
(up to today) at least

→ i.e. expansion of universe even when F_g is attractive

The equation is also similar to electric force

some may put a negative sign in front attractive

$$|\vec{F}| = -G \frac{m_1 \cdot m_2}{r^2}$$



where...

medium specific (although assume vacuum for all right now)

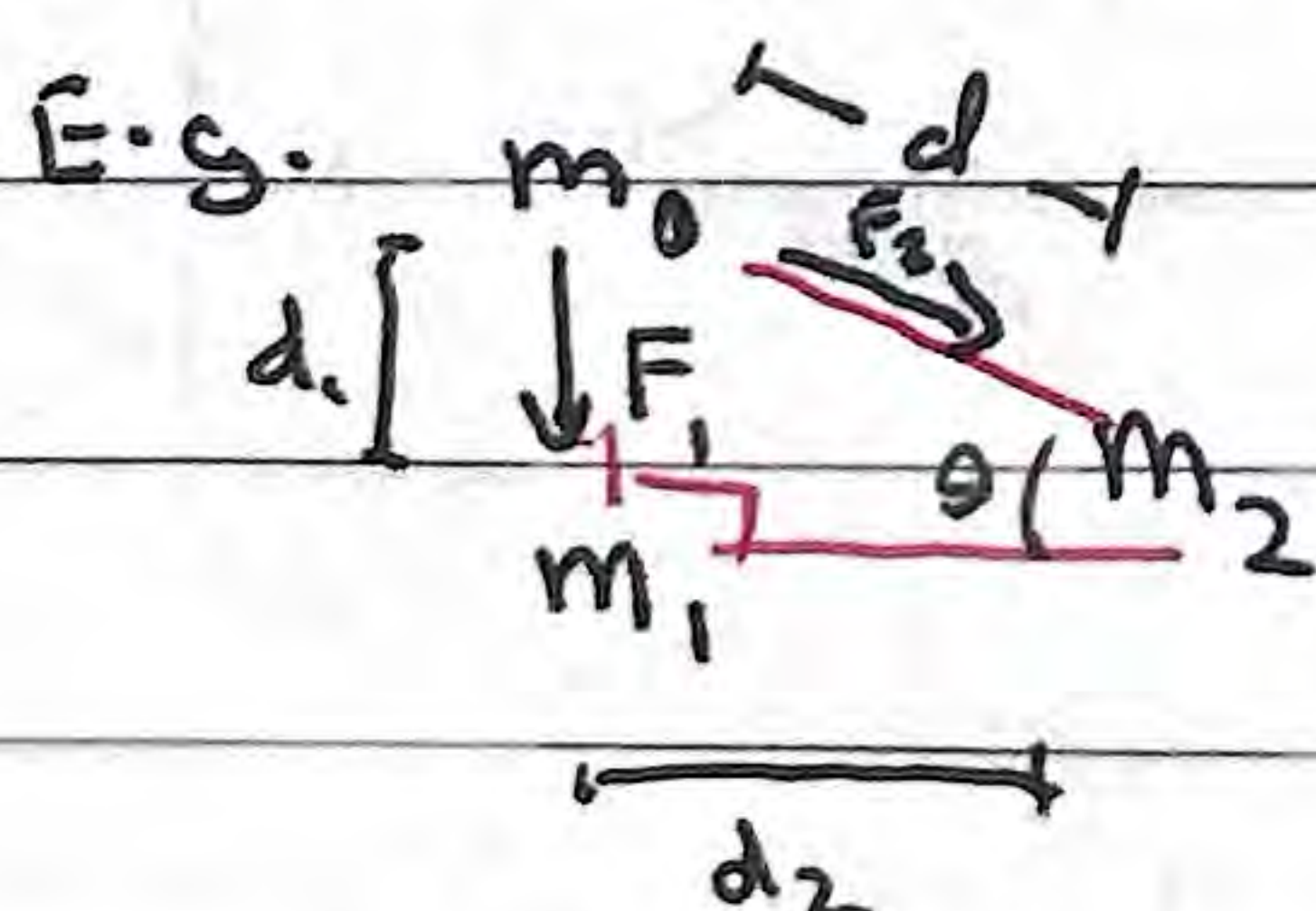
G - universal gravitational constant ($6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$)

m_n - mass of body n

r - distance between two bodies of mass

Gravitational force is a vector

- Thus to find the net force, it is necessary to decompose vectors



$$F_1 = \langle 0, -G \frac{m_1 m_0}{d_1^2} \rangle$$

$$F_2 = \langle G \frac{m_0 m_2}{d^2} \cdot \cos \theta, -G \frac{m_0 m_2}{d^2} \cdot \sin \theta \rangle$$

Gravitational Field Strength

The gravitational field strength at a point is the force per unit mass experienced by a point mass

similar to electric field strength...

$$g = \frac{F_g}{m}$$

the acceleration due to gravity is the gravitational electric field

and thus...

$$g = -\frac{G \cdot M}{r^2}$$

for earth!

$$= -9.8 \text{ N} \cdot \text{kg}^{-1} \quad (\text{ms}^{-2} = \text{N} \cdot \text{kg}^{-1})$$

where...

G - universal gravitational constant ($6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$)

M - mass of producing body

r - distance between two bodies of mass

Gravitational field is also a vector!

Gravitational Potential Energy

The gravitational potential energy of two masses is the work done (energy transferred) in bringing the two masses from infinity to their present location.

$$E_p = -G \cdot \frac{M \cdot m}{r}$$

external work = -
done by system is +

$$E_p = G \frac{M \cdot m}{r} = G \frac{M}{r} \cdot r \cdot m = G \frac{M}{r^2} r \cdot m = m \cdot g \cdot h$$

Gravitational potential Difference

The gravitational potential energy per unit mass

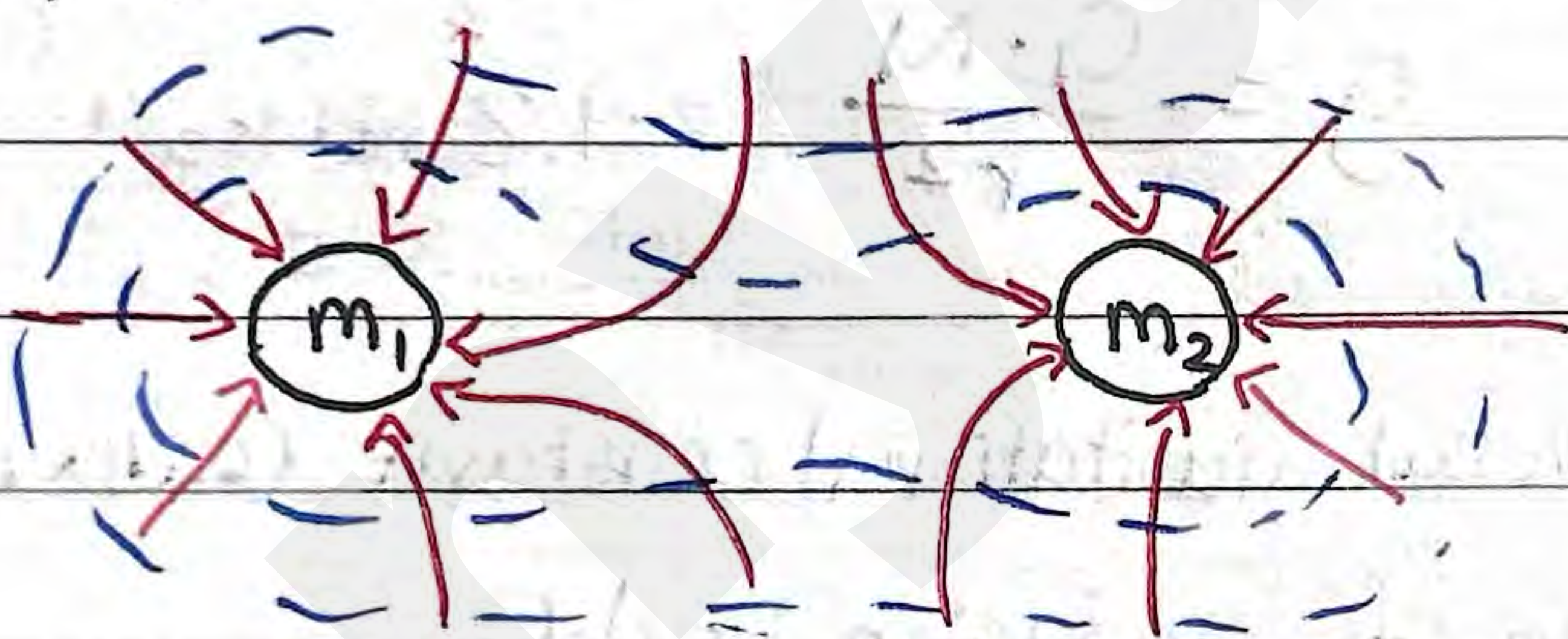
$$V_g = \frac{W}{m} = -\frac{GM}{r}$$

$$\downarrow$$

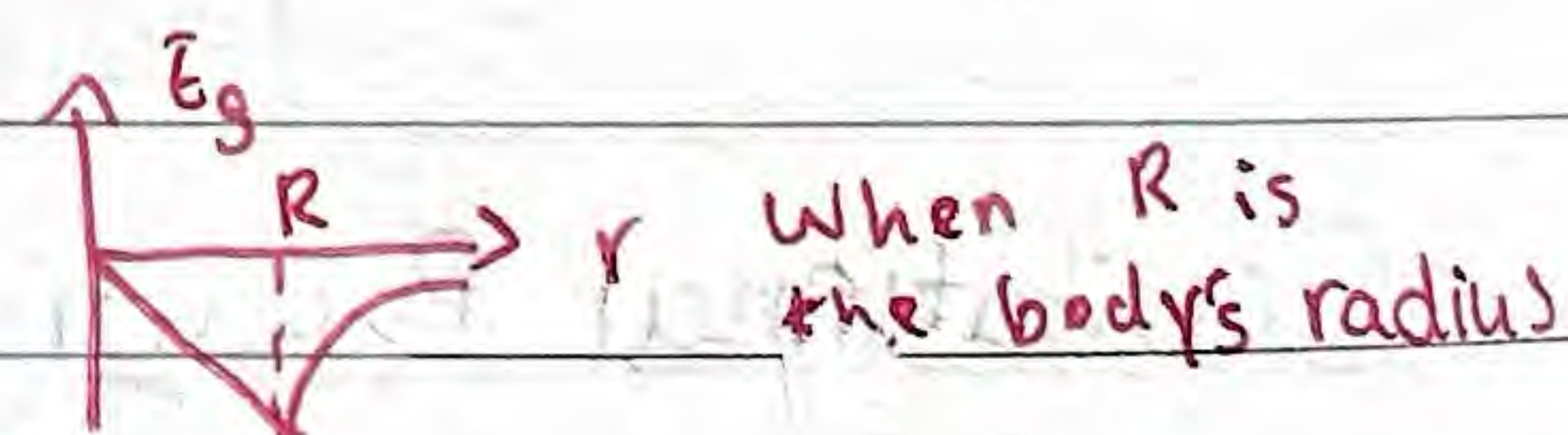
$$V_g = g \cdot h$$

Schematics

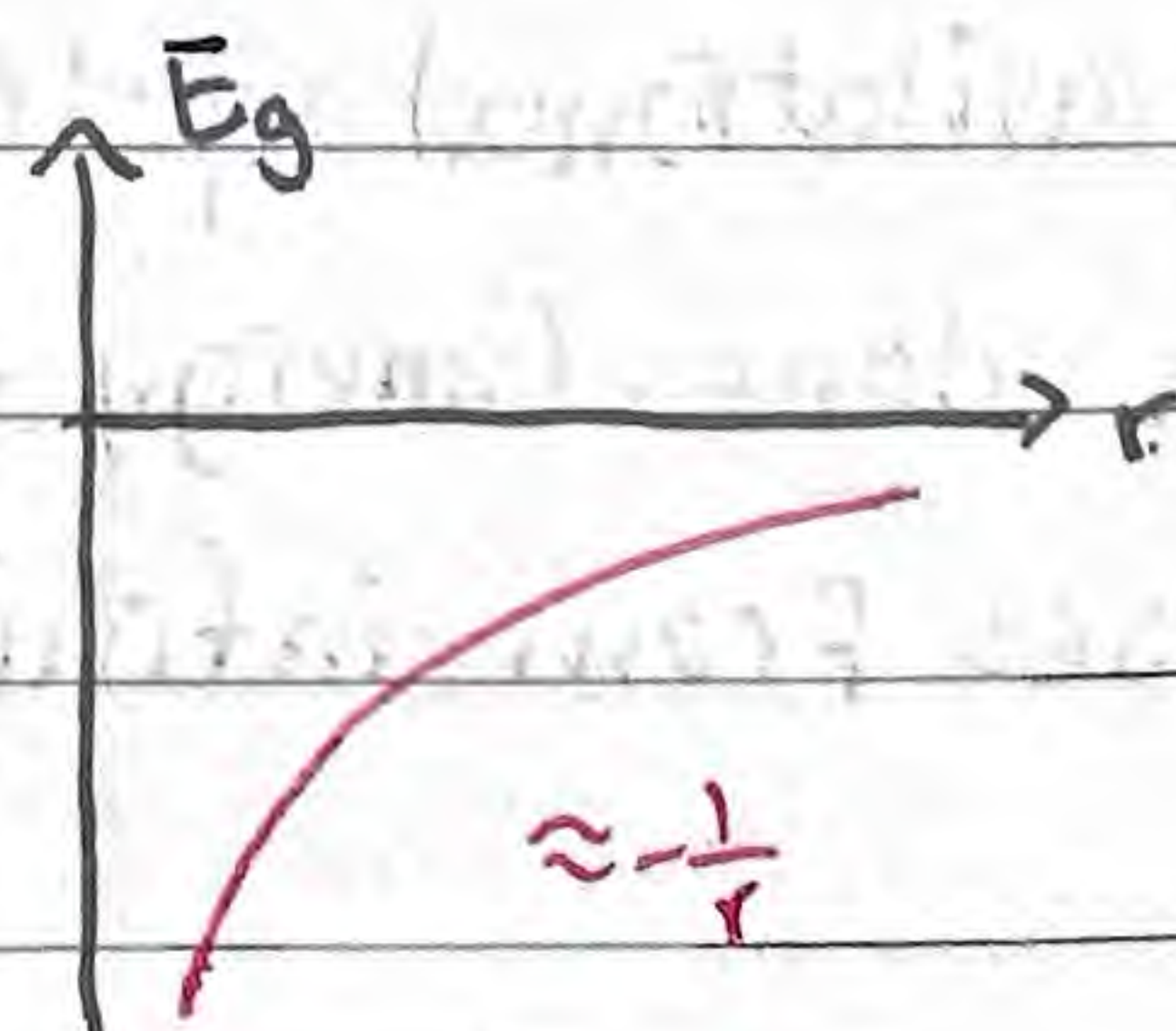
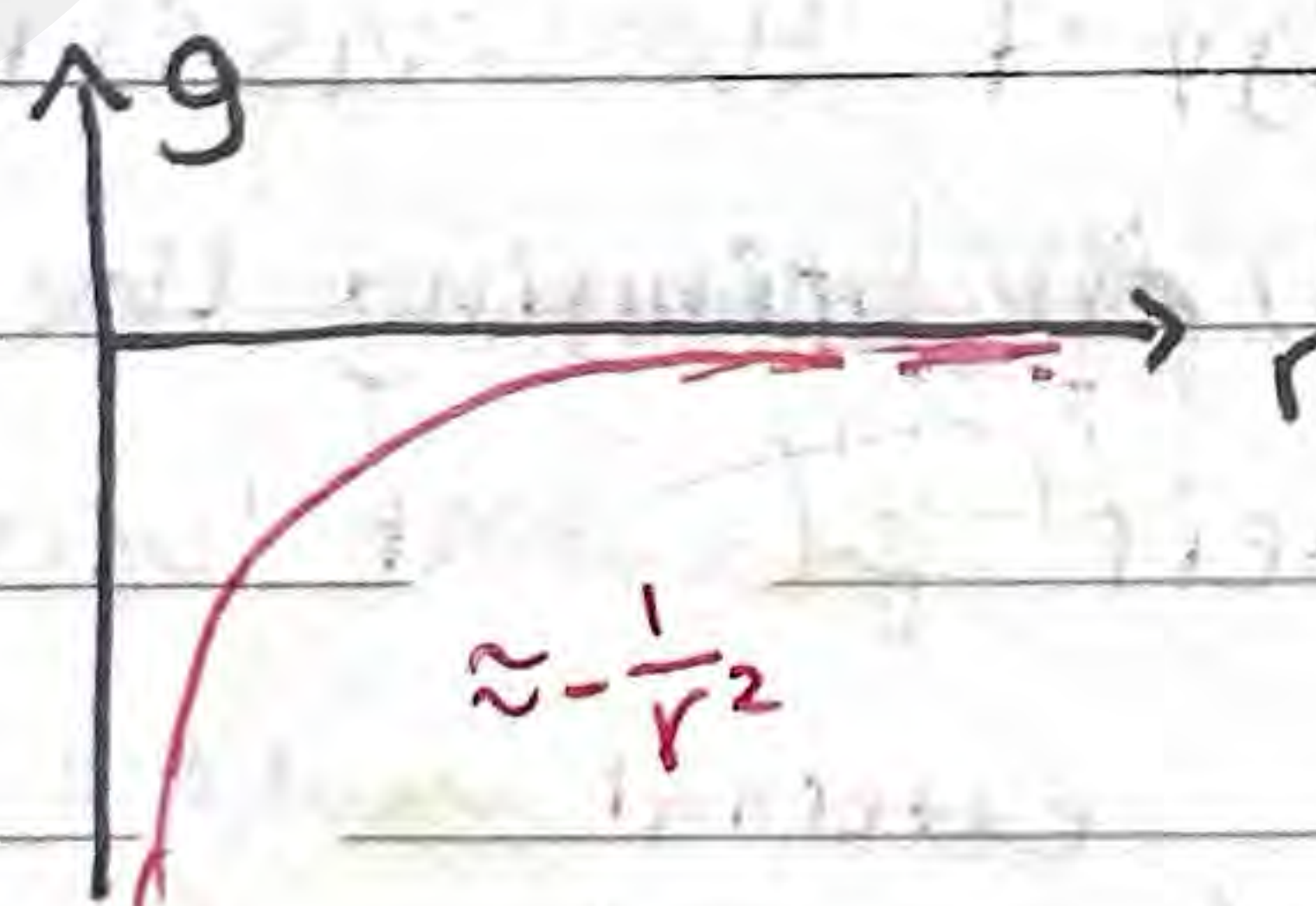
- We can draw gravitational field lines and equipotential lines similar to electrostatics



Remember to make equipotential lines further and further away



Graphs

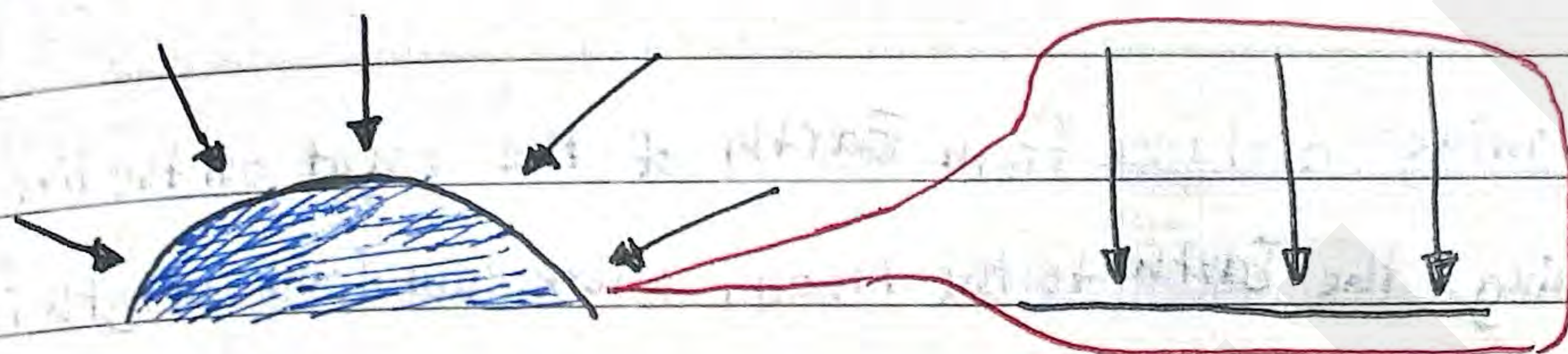


Gravitational Field

Gravitational potential energy

Uniform Gravitational Field

Because of how large some bodies of mass are, the surface of the body is assumed to be uniform in terms of the gravitational field



This why g is assumed to be a constant 9.81 m s^{-2}

* The further you are away from the center of the mass, the greater gravitational potential energy

Gravitational Potential problems

e.g. The mass of the Moon is about 81 times smaller than that of the Earth. The distance between the Earth and the Moon is about $d = 3.8 \times 10^8 \text{ m}$. The mass of the Earth is $5.97 \times 10^{24} \text{ kg}$.

a) Determine distance from Earth of the point on the line joining the Earth to the Moon, where net field strength is 0

$$g_E = g_M$$

$$+ \frac{GM}{x^2} = + \frac{Gm}{(d-x)^2}$$

$$\frac{M}{m} = \frac{x^2}{(x-d)^2}$$

$$9 = \frac{x}{d-x}$$

$$x = \frac{9}{10} d \approx 3.4 \times 10^8 \text{ m}$$

b) Combined gravitational potential at that point

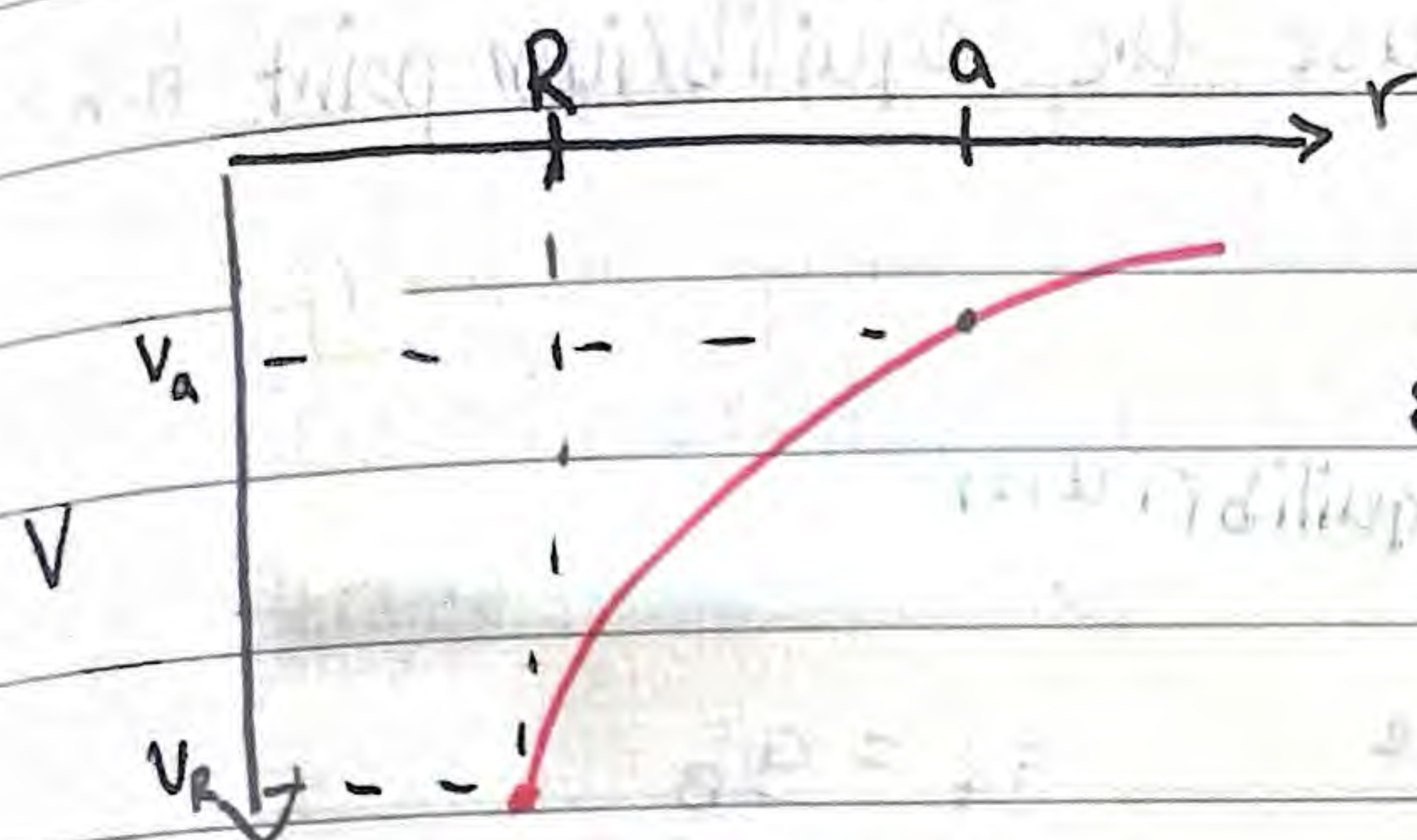
$$V_{g, \text{net}} = V_{g_1} + V_{g_2} = - \frac{GM}{d-x} = -1.3 \times 10^6 \text{ J kg}^{-1}$$

c) Work required to move 2500 kg to that point

$$2500 \times V_{g, \text{net}}$$

Gravitational Potential problemsWith graphs

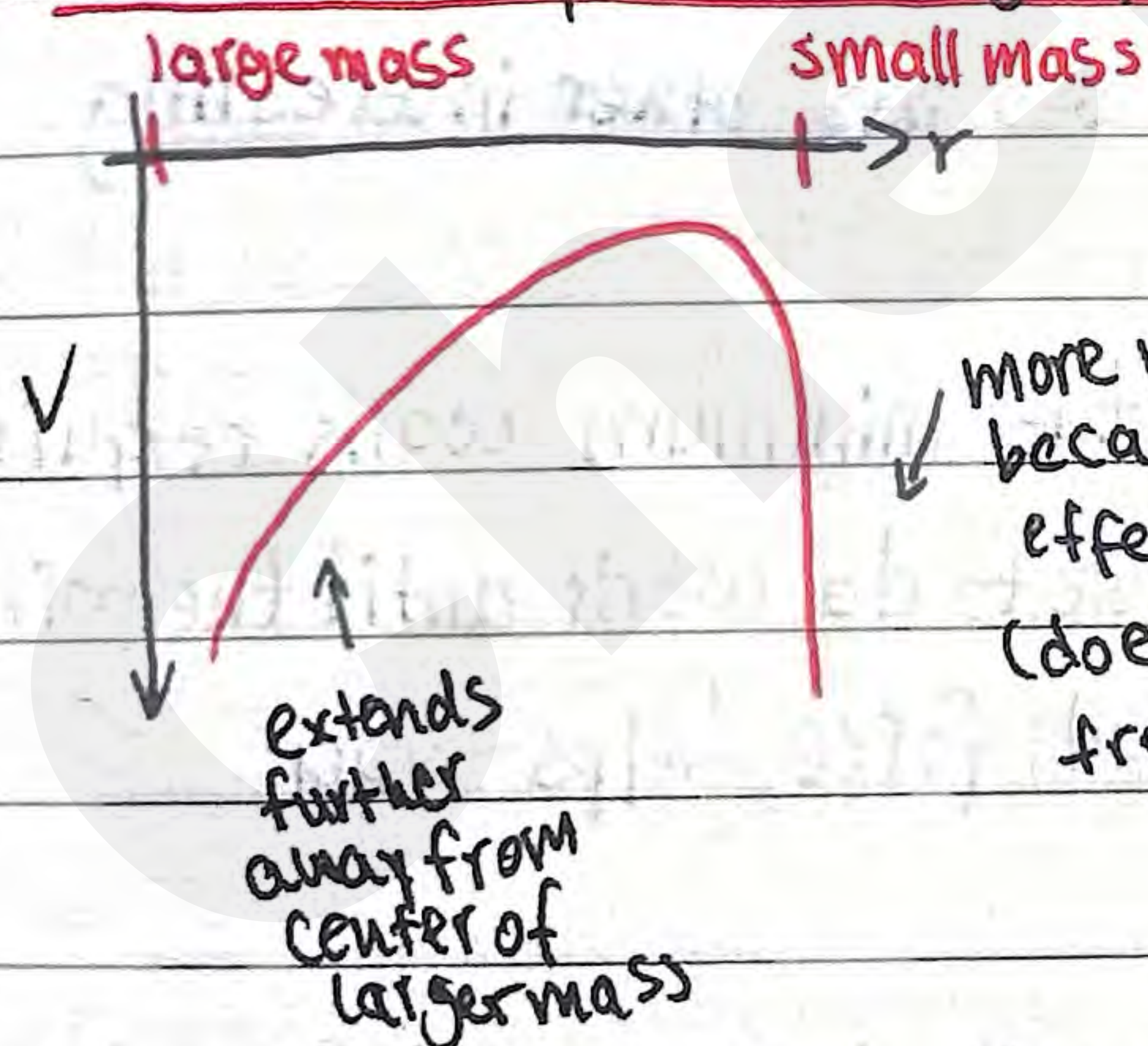
* Don't forget $W = \Delta V \cdot m$ not $W = V \cdot m$!



so work from R to a is
 $m \cdot (V_a - V_R)$

↳ since moving away center,
 work against system so
 it is expected to be > 0

Be aware of
 radius of mass
 (R)

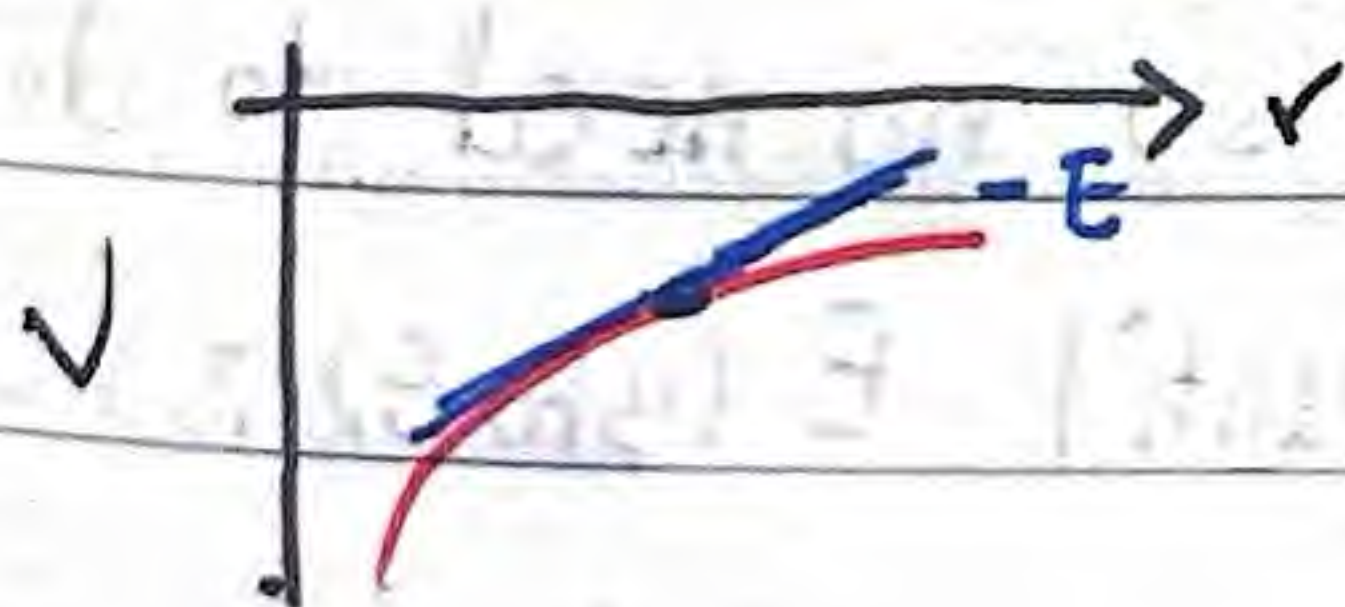
Gravitational potential graphs (two masses)

more vertical
 because weaker
 effect from smaller mass
 (does not extend far away
 from mass)

extends
 further
 away from
 center of
 larger mass

Gravitational field as slope

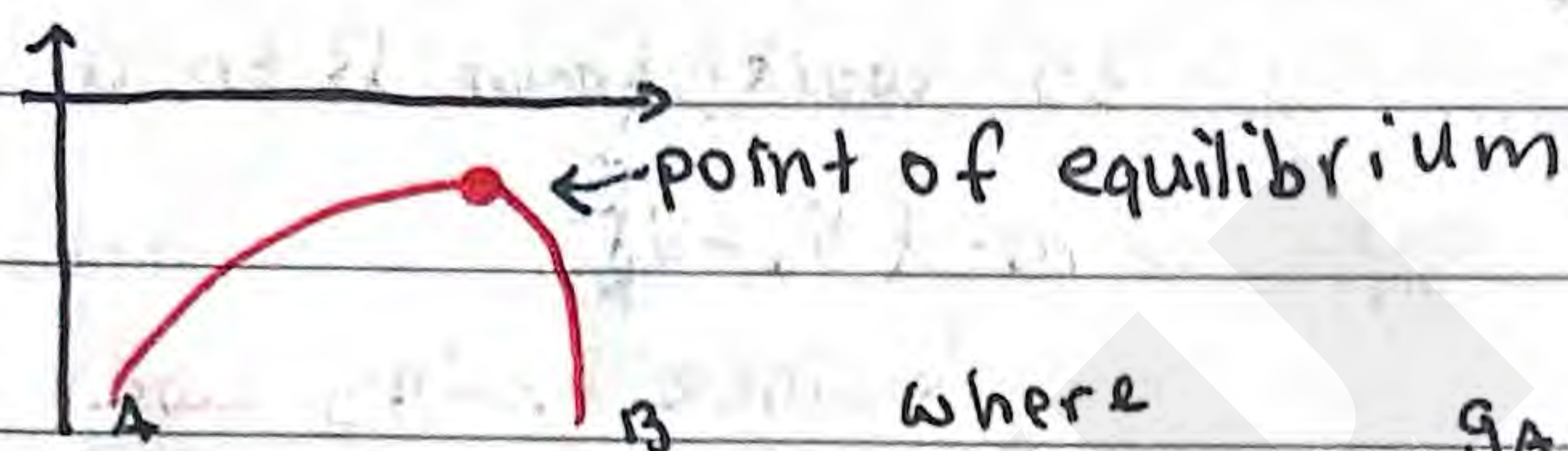
Recall $E = -\frac{dV}{dr}$ which also applies to gravitational $g = -\frac{dV}{dr}$



Gravitational Potential graphs

Sometimes, given graph, you may be asked to find the ratio between two masses

- The key here is to use the equilibrium point between the two masses



where

$$g_A = g_B$$

$$G \frac{A}{r_A^2} = G \frac{B}{r_B^2}$$

mass A →

→

$$\frac{A}{B} = \frac{r_A^2}{r_B^2}$$

← dist. of equilibrium from A

mass B →

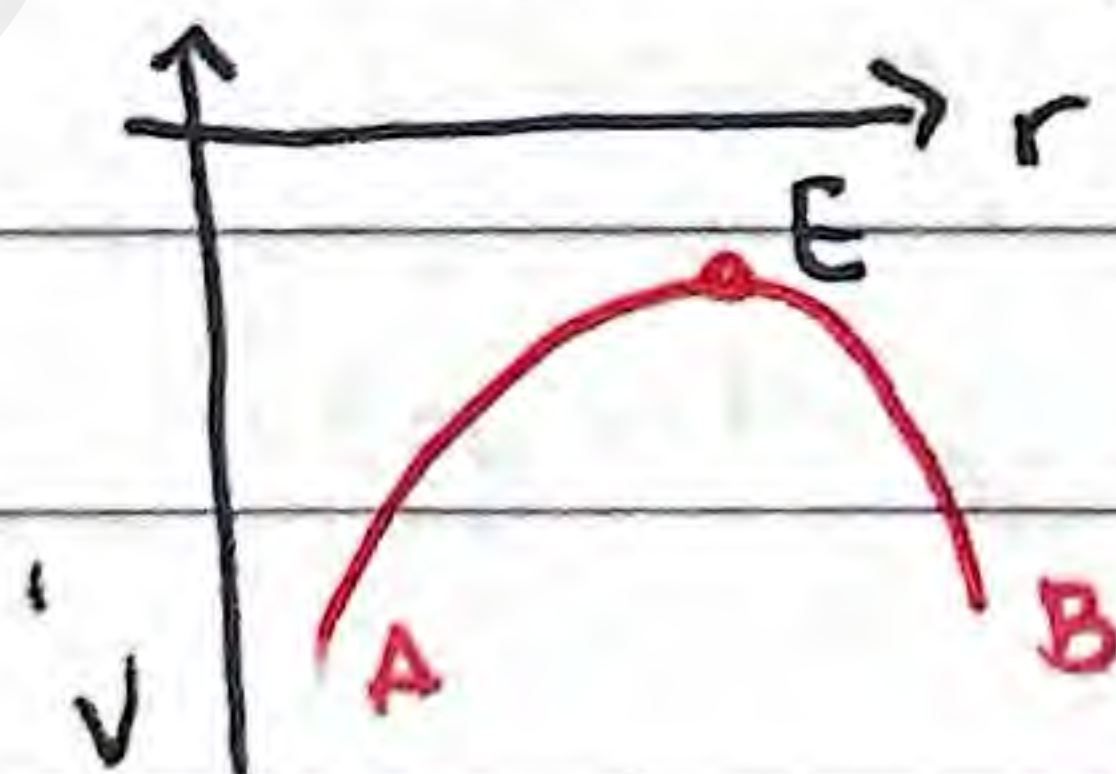
→

← dist. of equilibrium from B

The point of equilibrium also has other interesting properties

↳ It's useful to find the minimum work required

- Since you only have to do work until the point where gravitational force helps you



min work involves ΔV from A to E

From E to B if moving away from A, the gravitational force will act in its favor, so no need to do work

Similar with from B to A, only work until E (B to E) is necessary

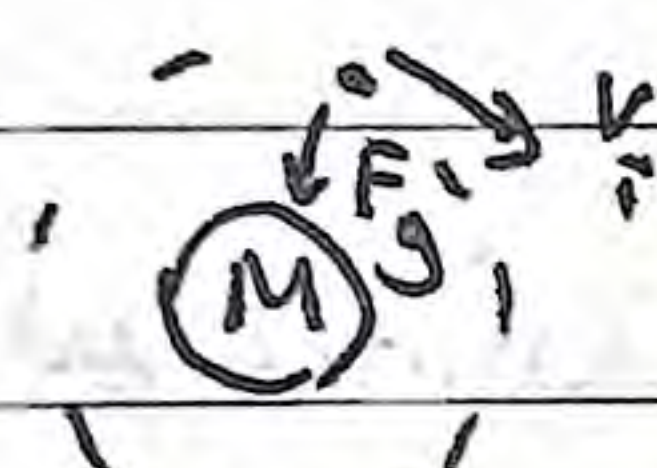
Orbital motion

connecting circular motion with gravitational force

Recall... $F_{net} = m \cdot a_c$

in an orbiting motion, the only force is F_g towards the larger mass's center

Thus



$$F_{net} = m \cdot a_c$$

$$G \frac{Mm}{r^2} = m \cdot \frac{v^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

← orbital speed
(speed, no more no less, to stay in same orbital radius)

independent of orbiting body's mass

You can also relate tangential velocity to period

$$v = \frac{\text{circumference}}{T} = \frac{2\pi r}{T}$$

← Kepler's 3rd law

$$T^2 = \frac{4\pi^2 r^2}{v^2} = 4\pi^2 r^2 \frac{r}{GM} = \frac{4\pi^2}{GM} \cdot r^3$$

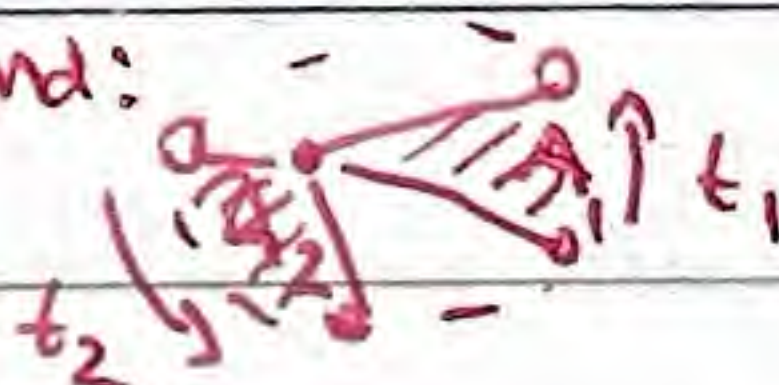
↑
Kepler's laws are not needed for IB

We can also derive another formula

$$\frac{4\pi^2}{GM} = \text{constant} = \frac{T^2}{r^3}$$

Kepler's first: orbit is elliptical (not circular)

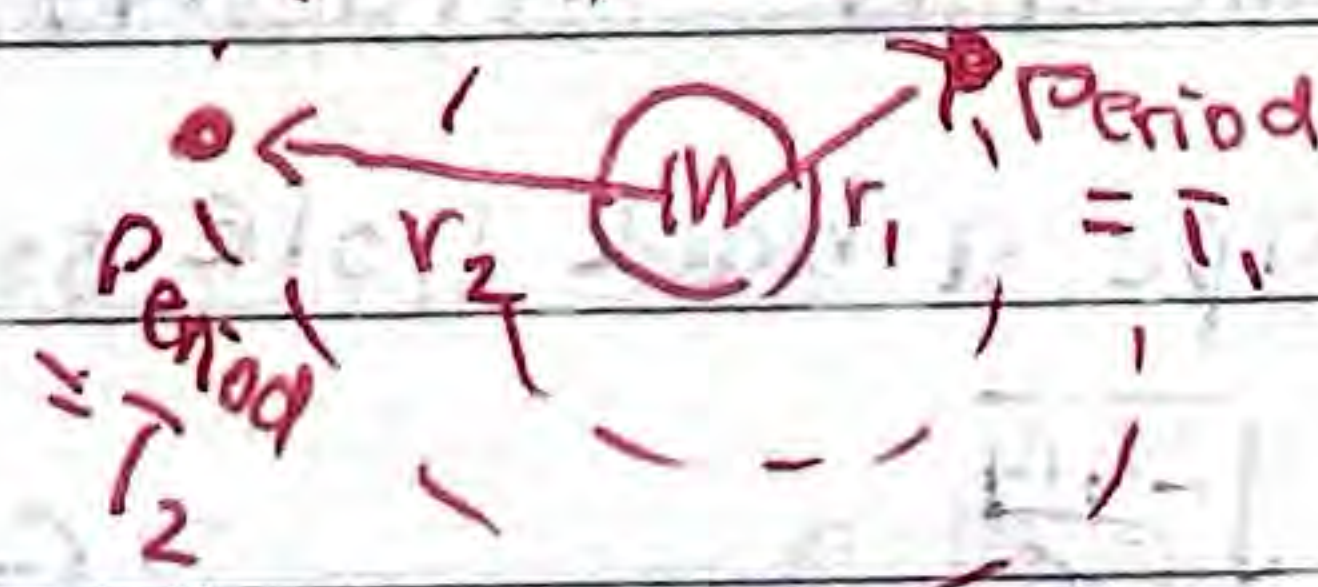
Kepler's second:



$$\frac{A_1}{A_2} = \frac{t_1}{t_2}$$

thus...

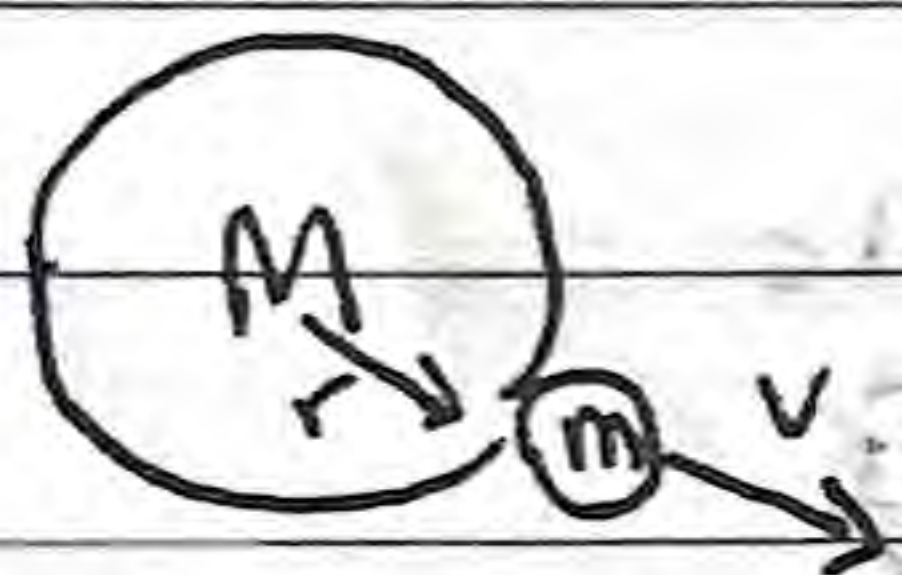
$$\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}$$



Escape velocity (speed)

Projectile - An object moving such that the only force affecting it is the gravitational force

The escape velocity of a projectile is the minimum initial velocity required so the projectile no longer falls down again $\Leftarrow$ It's a projectile so you can't apply a force to it so it leaves free fall (rockets are not projectiles)



analyze energy

initial: $E_{ki} > 0$ (initial velocity required)

$E_{pi} < 0$ ($r > 0$ and E_p is pointing down)

final: $E_{kf} = 0$ (min. velocity, E_{ki} just right)

$E_{pf} = 0$ ($E_p = 0$ ensures doesn't fall down)

Conservation of energy: $\frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right) = 0$

$$v_{\text{escape}} = \sqrt{\frac{2GM}{r}}$$

So for earth, $v_{\text{esc}} \approx 11,000 \text{ m s}^{-1}$

We can also think of it differently

also $c^2 = \frac{2GM}{r}$

$m \cdot c^2 = \frac{2GM}{r} m$

$E = mc^2$

light can't escape a black hole so we can set $v_{\text{esc}} = c$

$$c = \sqrt{\frac{2GM}{r}}$$

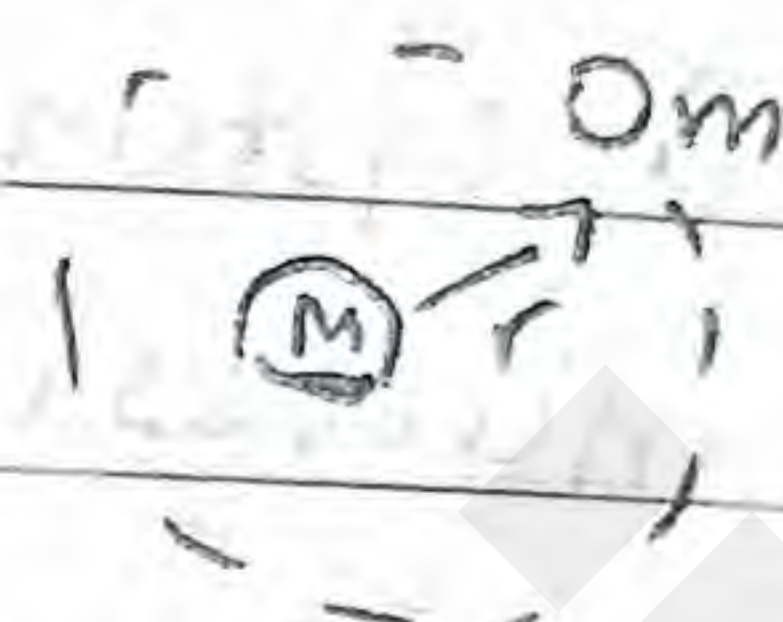
event horizon of mass M black hole

$$\rightarrow r = \frac{2GM}{c^2}$$

Schwarzschild radius

orbital energies

• Total energy of orbital motion



$$E_T = E_K + E_P$$

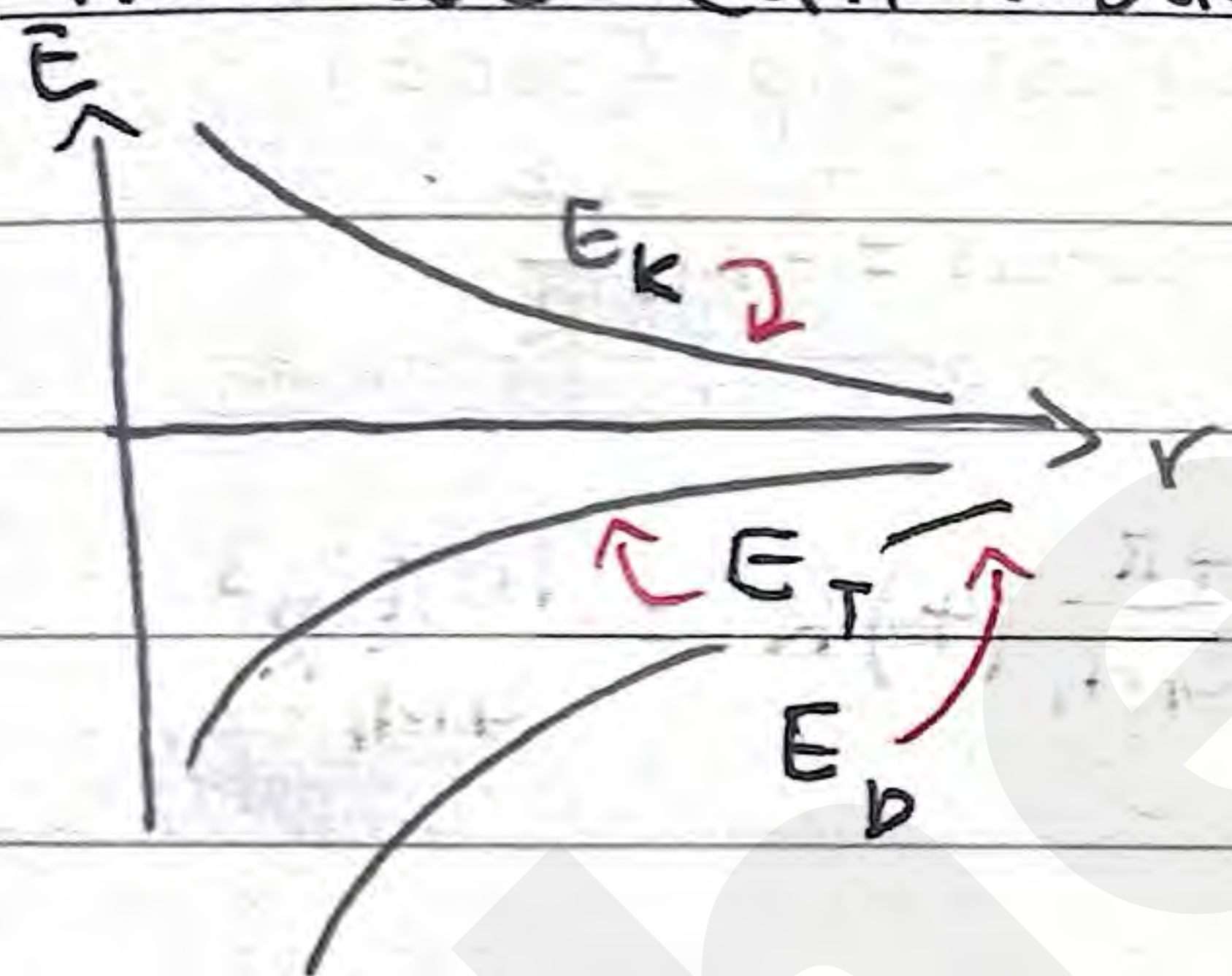
$$\begin{aligned} E_T &= \frac{1}{2}mv^2 - \frac{GMm}{r} \\ &= \frac{1}{2}m\left(\frac{GM}{r}\right) - \frac{GMm}{r} \\ &= -\frac{1}{2}\frac{GMm}{r} \end{aligned}$$

Recall

$$E_K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{GM}{r}\right)$$

$$E_P = -\frac{GMm}{r}$$

Thus we can visualize energy vs radius graph



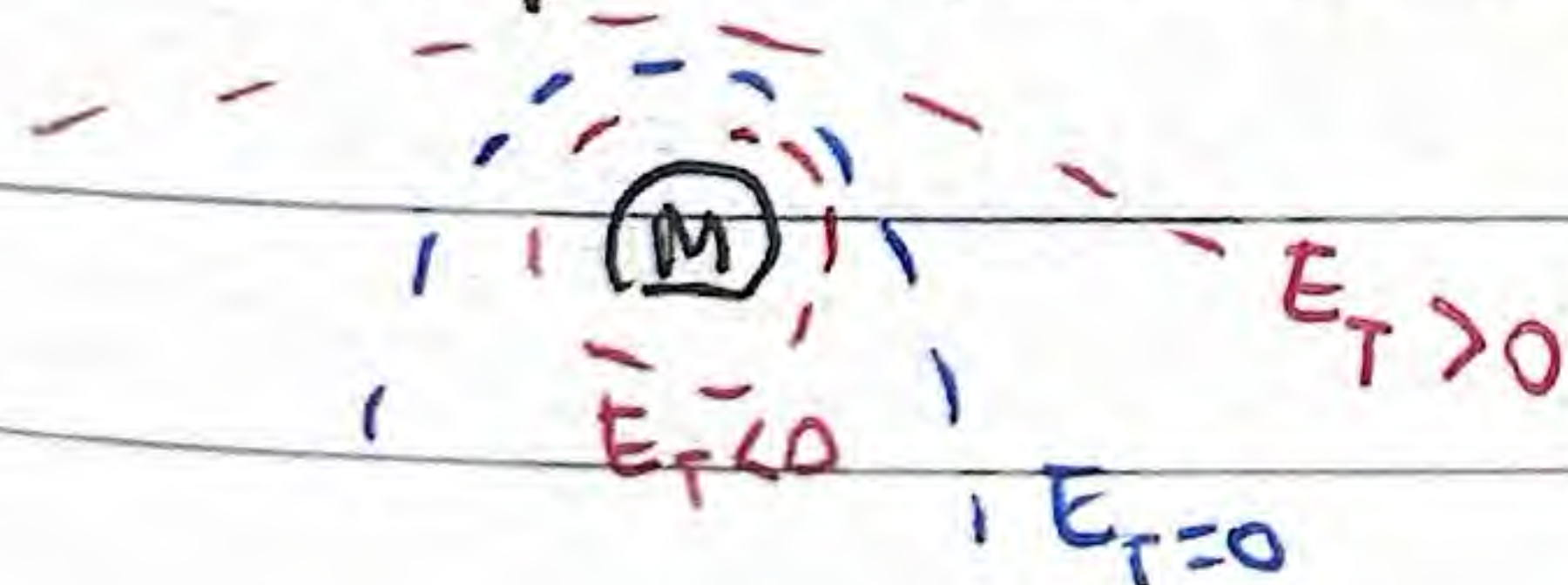
* E_T and E_K are mirrored graphs

The signs of E_T can also tell us a lot

$E_T < 0$, orbit of mass m is circular or elliptic

$E_T = 0$, orbit of mass m is parabolic

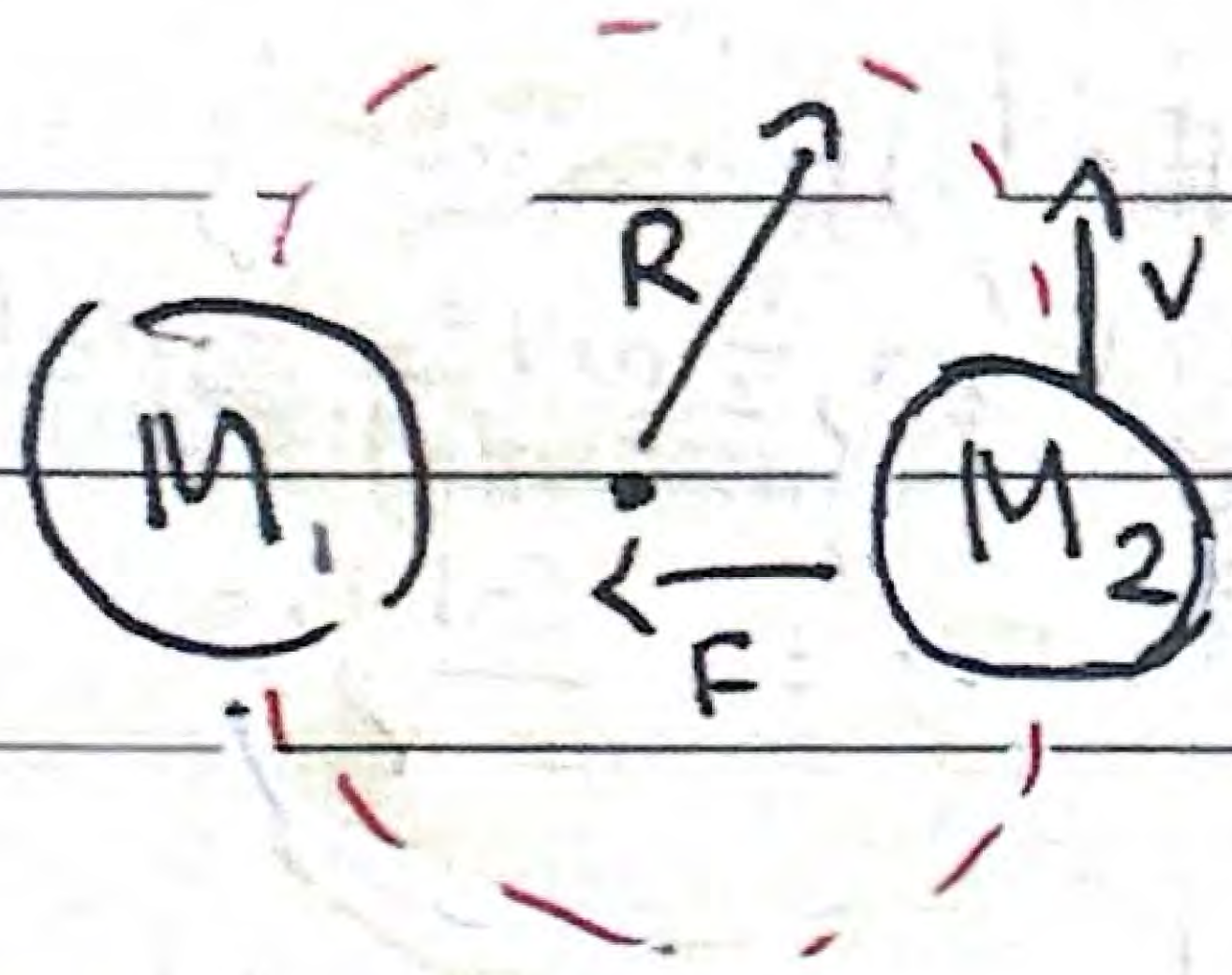
$E_T > 0$, orbit of mass m is hyperbolic



Binary System orbital motion

A system in which two bodies orbit each other such that their motions are affected by each other

↳ Kepler's 3rd law does not apply here



M_2 is affected by M_1 's pull and vice versa

$F_{\text{grav}} = -G \frac{Mm}{r^2}$ where $r = 2R$ (dist. between 2 masses)

$$\frac{-GM_1 M_2}{4R^2} = -M_2 \frac{v^2}{R} \quad \leftarrow \text{Newton's 3rd law, equal opp. forces}$$

$\uparrow$ $F_{\text{grav centripetal}} = -M_2 \frac{v^2}{R}$

$$v^2 = \frac{GM_1}{4R} \Rightarrow T^2 = \frac{4\pi r^2}{v^2} = \frac{4\pi}{GM_1} (4) R^3 = \frac{16\pi}{GM_1} R^3$$